QUADRATIC YOUNG INEQUALITIES ON DISCRETE ALPHABETS: EXACT SMALL MODELS AND MULTISCALE RIGIDITY

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Abstract. For the binary cube $\{0,1\}^d$, we determine the exact best constant in the off-diagonal quadratic Young inequality

$$\|f * g\|_{\ell^2(\mathbb{Z}^d)} \le C\|f\|_{\ell^p(\mathbb{Z}^d)}\|g\|_{\ell^q(\mathbb{Z}^d)}$$

for every $p, q \in [1,\infty]$ and every dimension d, and hence obtain the complete dimension-free exponent region together with sharp cross-energy, sumset, entropy, and Fourier-overlap consequences. For the ternary cube $\{0,1,2\}^d$, we determine the sharp dimension-free L^4 exponent and the complete one-dimensional best-constant curve; every nontrivial nonnegative extremizer is reflection symmetric and center heavy. For arbitrary finite interval alphabets, we prove arithmetic-progression support rigidity for exact extremizers, a universal one-third law for the reflection-odd Hessian, the universal reflection-stability threshold $p = 4/3$, an exact parity-renormalization formula with a nonnegative square defect, an exact multiplicative cascade, and discrete–continuous and entropy identities that isolate the large-alphabet problem. The predicted Gaussian first-order large-alphabet asymptotic remains conditional on a stated multiscale refinement-rigidity conjecture. Finally, for output exponents $r > 2$ sufficiently close to 2, we determine sharp biased–biased equality branches for fixed $q \in (1, 4/3)$ and the global flat-to-biased transition near the upper endpoint.

Contents

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1. Introduction

1.1. The common problem. For an integer $m \geq 2$, let

$$I_m := \{0, 1, \ldots, m-1\} \subset \mathbb{Z}.$$

The basic object of this paper is the support-restricted Young quotient

$$\mathsf{C}_{m,d}(p, q; r) := \sup_{\substack{f,g \neq 0 \\ \operatorname{supp} f, \operatorname{supp} g \subset I_m^d}} \frac{\|f * g\|_{\ell^r(\mathbb{Z}^d)}}{\|f\|_{\ell^p(\mathbb{Z}^d)} \|g\|_{\ell^q(\mathbb{Z}^d)}}. \tag{1.1}$$

Without the support restriction, the classical Young inequality has constant one on

$$1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}.$$

A finite product alphabet changes the geometry: in each coordinate only finitely many translates can interact, and one may retain dimension-independent constants beyond the classical Young plane. The central questions are therefore to determine the exact constant, the constant-one exponent region, and the geometry of extremizers.

The quadratic output $r = 2$ has a second interpretation. For a finitely supported $f : \mathbb{Z}^d \to \mathbb{C}$, write

$$\widehat{f}(\xi) = \sum_{x \in \mathbb{Z}^d} f(x) e^{-2\pi i x \cdot \xi}, \qquad \xi \in \mathbb{T}^d,$$

where Haar measure on $\mathbb{T}^d$ is normalized. Plancherel gives

$$\|\widehat{f}\|_{L^4(\mathbb{T}^d)}^4 = \|f * f\|_{\ell^2(\mathbb{Z}^d)}^2. \tag{1.2}$$

For $f = \mathbf{1}_A$, this common quantity is the additive energy

$$E(A) := \#\{(a_1, a_2, a_3, a_4) \in A^4 : a_1 + a_2 = a_3 + a_4\}. \tag{1.3}$$

Thus the self-convolution problem can be read interchangeably as a sharp L^4 Fourier inequality or a dimension-free additive-energy inequality. The bilinear quotient $\|f * g\|_2$ similarly measures cross-additive energy and, after normalization to probability masses, collision entropy.

The paper follows the increase in alphabet complexity

$$I_2 \longrightarrow I_3 \longrightarrow I_m.$$

The binary model is completely solvable even off the diagonal: exact tensorization reduces the problem to two scalar balance variables, and the Hilbert-space structure at $r = 2$ forces the extremum to the boundary. The ternary model is the first one with a genuine non-flat interior profile: the sharp self-convolution extremizer is center heavy. For a general alphabet, direct finite-dimensional optimization is replaced by variational, reflection, and multiscale structure. The exact results lead to a two-well picture in which near extremizers are either atomic or broad and approximately Gaussian.

1.2. Background. The sharp Euclidean constants in Hausdorff–Young and Young inequalities were determined by Beckner and by Brascamp–Lieb [1, 6]. On product sets, the reverse or support-size side of convolution has a long combinatorial history. Woodall and, independently, Hajela and Seymour proved the sharp binary-cube Brunn–Minkowski inequality

$$|A + B| \geq (|A||B|)^{\log_2 3/2}, \qquad A, B \subset \{0, 1\}^d,$$

while dimension-free sumset estimates for more general product sets appeared in work of Bourgain, Dilworth, Ford, Konyagin, and Kutzarova [16, 9, 5]. Becker, Ivanisvili, Krachun, and Madrid later established the sharp ternary analogue and developed a functional sup-convolution formulation [2]. In the limit $r \downarrow 0$, reverse Young inequalities recover this support-size theory; see Leindler and Beltran–Ivanisvili–Madrid–Patil [11, 4].

Positive output exponents measure representation multiplicity rather than only support. On the binary cube, Kane and Tao proved the sharp additive-energy exponent, and de Dios Pont, Greenfeld, Ivanisvili, and Madrid developed higher-energy inequalities [10, 8]. Generalized additive energies in arbitrary abelian groups were developed systematically by Shkredov: [15] studies quantities such as E_k and T_k, their relations with Gowers norms, and the additive structure forced by critical relations among these energies. Related higher-moment convolution estimates appear in work of Schoen and Shkredov [13]. Beker, Crmarić, and Kovač placed cube inequalities in a broader Gowers-norm framework and proved the dimension-compression principle used in the ternary part of the present paper [3]. Shao obtained nontrivial upper and lower asymptotic bounds for interval alphabets of growing size [14].

The recent works of Beltran, Ivanisvili, Madrid, and Patil and of Crmarić, Kovač, and Shiraki initiated a systematic sharp functional theory on the binary cube [4, 7]. In particular, for every $r \geq 1$ they determined the diagonal exponent

$$p_r = \frac{2r}{\log_2(2+2^r)}$$

for which $\|f * g\|_r \leq \|f\|_{p_r}\|g\|_{p_r}$ holds in every dimension. Beltran, Ivanisvili, Madrid, and Patil also obtained necessary conditions and partial off-diagonal results at $r = 2$. A key local obstruction was recorded by Crmarić, Kovač, and Shiraki in [7, Remark 15]: on the full-cube critical line, the flat–flat profile ceases to be a local maximum as soon as $p < 4/3$ or $q < 4/3$. They also indicated how the complementary flat segment can be approached through endpoint verification and interpolation. Thus previous work identified both the full-cube regime and the onset of symmetry breaking, but did not determine the profile replacing the flat point or the complete best constant. The binary analysis below gives an independent complete solution at quadratic output: a global curvature argument forces an extremizer to an edge, where the missing boundary is described by a flat–biased one-variable curve. The final sections explain why $r = 2$ is a special, Hilbertian point and how the equality geometry changes immediately for $r > 2$.

Independent forthcoming work. José Madrid and Paata Ivanisvili informed the author that they and their collaborators had independently solved the full binary $r = 2$ problem—the original problem addressed in their forthcoming manuscript—before receiving an earlier draft of the present paper. Based on these private communications, the author understands that the overlap concerns the binary $r = 2$ phase diagram; the ternary, general-alphabet, and transverse r-directions developed here are separate. At the time of writing, no public manuscript is available to cite. Their preprint is expected to appear on arXiv shortly; when it becomes available, the author will add an appropriate citation and a fuller comparison in a subsequent version.[1]

1.3. **The binary off-diagonal theorem.** Put $Q_d := \{0,1\}^d$. For $1 \leq q < \infty$, define

$$\Lambda(q) := \max_{0 \leq t \leq 1} \log_2 \frac{1+t+t^2}{(1+t^q)^{2/q}}, \tag{1.4}$$

set $\Lambda(\infty) := \log_2 3$, and put

$$\mathsf{M}(p,q) := \max\left\{0,\ 1-\frac{2}{p}+\Lambda(q),\ 1-\frac{2}{q}+\Lambda(p)\right\}. \tag{1.5}$$

As usual, $1/\infty = 0$.

Theorem 1.1. *Let $d \geq 1$ and $p, q \in [1,\infty]$. Then*

$$\|f * g\|_2 \leq 2^{\frac{d}{2}\mathsf{M}(p,q)}\|f\|_p\|g\|_q \tag{1.6}$$

[1] Private communications, August 2026.

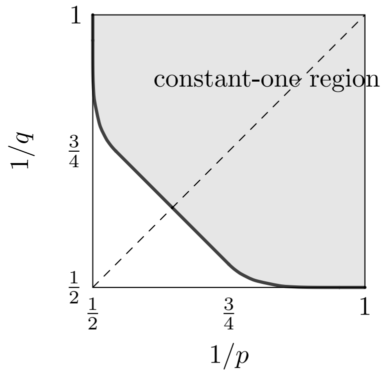

FIGURE 1. The exact constant-one region for quadratic output on the binary cube. The curved pieces correspond to flat–biased one-dimensional extremizers; the central straight pieces lie on the full-cube line.

for all $f, g\colon \mathbb{Z}^d \to \mathbb{C}$ *supported on* Q_d*. The constant is optimal.*

Corollary 1.2. *The estimate*

$$\|f * g\|_2 \le \|f\|_p \|g\|_q \tag{1.7}$$

holds in every dimension for all f, g *supported on* Q_d *if and only if*

$$\frac{1}{p} \ge \frac{1+\Lambda(q)}{2}, \qquad \frac{1}{q} \ge \frac{1+\Lambda(p)}{2}. \tag{1.8}$$

If $p \ge q$*, the second inequality follows from the first and the sharp boundary is*

$$\frac{1}{p} = \frac{1+\Lambda(q)}{2}. \tag{1.9}$$

The one-variable maximum has a transition at $q = 4/3$:

$$\Lambda(1) = 0, \qquad \Lambda(q) = \log_2 3 - \frac{2}{q} \quad (q \ge 4/3). \tag{1.10}$$

For $1 < q < 4/3$, the unique maximizing parameter $t_q \in (0, 1)$ satisfies

$$t_q^{q-1}(2 + t_q) = 1 + 2t_q. \tag{1.11}$$

The curved boundary joins $(1/2, 1)$ to $(\log_2(3)/2 - 1/4, 3/4)$ in the coordinates $(1/p, 1/q)$; it then meets the full-cube line and its reflection. The resulting constant-one region is shown in Figure 1.

Remark 1.3. The full-cube portion of Figure 1 was anticipated by the calculation in [7, Remark 15]. On the critical line

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{2}\log_2 6,$$

Crmarić, Kovač, and Shiraki computed the Hessian at the flat–flat profile and observed that it cannot be a local maximum when $p < 4/3$ or $q < 4/3$; they also indicated an endpoint-and-interpolation route for the complementary flat regime. The curved boundary in Corollary 1.2 is the further phenomenon forced by this instability: below the flat threshold, the sharp finite profile moves to a flat–biased edge. The global edge reduction in Proposition 4.1 proves this displacement and determines the exact profile and best constant.

For indicators, Theorem 1.1 gives the complete asymmetric cross-energy region. Cauchy–Schwarz then yields an asymmetric family of sumset estimates. The same theorem is equivalently a sharp collision-entropy inequality and a bilinear Fourier-overlap inequality; these consequences are recorded after the analytic proof.

1.4. **The ternary model.** For a one-dimensional profile supported on I_n, set

$$Q(f) := \|f * f\|^2_{\ell^2(\mathbb{Z})}, \tag{1.12}$$

and

$$\mathcal{C}_n(p) := \sup_{0 \neq f\colon I_n \to \mathbb{C}} \frac{Q(f)^{1/4}}{\|f\|_{\ell^p(I_n)}}. \tag{1.13}$$

For the ternary alphabet, let x_* be the unique zero in $(3, 31/10)$ of the function Ψ defined in (8.6); define q_*, p_*, t_* by (8.7).

Theorem 1.4 (Sharp ternary constant-one threshold)**.** *One has*

$$p_* = 1.470203929717889502\ldots, \qquad t_* := \frac{4}{p_*} = 2.720710997397171381\ldots.$$

Every $f\colon \mathbb{Z}^d \to \mathbb{C}$ supported in $\{0,1,2\}^d$ satisfies

$$\|\widehat{f}\|_{L^4(\mathbb{T}^d)} \leq \|f\|_{\ell^p(\mathbb{Z}^d)}$$

for every d if and only if $p \leq p_*$. At $p = p_*$ a nonconstant one-dimensional equality profile is proportional to

$$(1, \sqrt{x_*}, 1), \qquad x_* = 3.066892946956241896\ldots.$$

Consequently,

$$E(A) \leq |A|^{t_*} \qquad (A \subset \{0,1,2\}^d),$$

and t_* is best possible uniformly in d.

The uniform profile is not extremal: the first genuinely interior small-alphabet optimizer is reflection symmetric but center heavy. The complete one-dimensional curve is as follows.

Theorem 1.5 (Complete ternary best-constant curve)**.** *For every $p > 0$,*

$$\mathcal{C}_3(p)^4 = \sup_{x \geq 0} \frac{x^2 + 12x + 6}{(x^{p/2} + 2)^{4/p}}. \tag{1.14}$$

For $0 < p \leq p_*$ the value is 1. For $p > p_*$ there is a unique $x_p \in (1, x_*)$ satisfying

$$x_p^{1-p/2}(x_p + 6) = 3(x_p + 1), \tag{1.15}$$

and, up to scalar multiplication and reflection, the unique nonnegative maximizer is $(1, \sqrt{x_p}, 1)$. Moreover,

$$\mathcal{C}_3(p)^4 = \frac{[3(x_p + 1)]^{4/p}}{(x_p^2 + 12x_p + 6)^{4/p - 1}} \qquad (p > p_*). \tag{1.16}$$

In particular,

$$\mathcal{C}_3(2)^4 = \frac{15}{7}, \qquad \mathcal{C}_3(4)^4 = 2 + \sqrt{19}, \qquad \lim_{p \to \infty} \mathcal{C}_3(p)^4 = 19.$$

For $0 < p \leq 4$, the one-coordinate ternary constant tensorizes exactly. Thus the one-variable formula also determines the sharp d-dimensional norm in the range relevant to the L^4 problem.

1.5. **General alphabets: reflection and renormalization.** Let $N = n-1$ and let J be reflection on I_n,

$$(Jf)_j = f_{N-j}.$$

For a positive symmetric profile $s = Js$, define

$$R_s(k) := \sum_j s_j s_{j+k}, \qquad C_s(i) := (s * s * \widetilde{s})(i),$$

and

$$P_s(i,j) := \frac{R_s(i-j)s_j}{C_s(i)}, \qquad \pi_i := \frac{s_i C_s(i)}{Q(s)}. \tag{1.17}$$

The kernel P_s is positive and reversible on $L^2(\pi)$.

Theorem 1.6 (Universal one-third law)**.** *For every reflection-odd function ϕ, meaning $\phi_{N-j} = -\phi_j$,*

$$0 \le \langle \phi, P_s\phi\rangle_{L^2(\pi)} \le \frac{1}{3}\|\phi\|^2_{L^2(\pi)}. \tag{1.18}$$

The upper endpoint is attained by $\phi_j = j - N/2$. If s has full positive support, this is the only odd eigenfunction at eigenvalue $1/3$, up to a scalar.

At a positive symmetric critical point of

$$\Phi_p(f) := \frac{Q(f)}{\|f\|_p^4},$$

the stationary measure is normalized p-mass, and the exact second variation yields

$$\frac{\Phi_p''(s)[s\phi, s\phi]}{4\Phi_p(s)} \le -\left(p - \frac{4}{3}\right)\|\phi\|^2_{L^2(\pi)}. \tag{1.19}$$

Hence $p = 4/3$ is the universal reflection-stability threshold. The same variational framework proves that the support of every nontrivial exact extremizer is an arithmetic progression; after translation and integer dilation, boundary extremizers are therefore full-support extremizers on smaller alphabets.

For the multiscale analysis, split a sequence into parity children

$$e_j = f_{2j}, \qquad o_j = f_{2j+1},$$

and set $x = Q(e)^{1/4}$, $y = Q(o)^{1/4}$.

Theorem 1.7 (Exact one-step renormalization)**.** *There is an explicitly defined defect $\Delta_{\mathrm{ren}}(e,o) \ge 0$ such that*

$$Q(f) = x^4 + y^4 + \big(6 - \Delta_{\mathrm{ren}}(e,o)\big)x^2y^2. \tag{1.20}$$

If $E = \widehat{e}$, $O = \widehat{o}$, and $O_{1/2}(\xi) = e^{-\pi i\xi}O(\xi)$, then

$$\begin{aligned}\Delta_{\mathrm{ren}}(e,o) = 3\left\|\frac{|E|^2}{x^2} - \frac{|O|^2}{y^2}\right\|^2_{L^2(\mathbb{T})} \\ + \frac{\|E\overline{O_{1/2}} - \overline{E}\,O_{1/2}\|^2_{L^2(\mathbb{T})}}{x^2y^2}.\end{aligned} \tag{1.21}$$

Iterating the identity gives an exact multiplicative cascade for $Q(f)^{p/4}$.

The square representation says that a nearly lossless balanced split has both magnitude and half-lattice phase compatibility. At the Euclidean exponent $p = 4/3$, a one-scale inverse statement forces every nearly lossless split to be either almost atomic or almost balanced with small defect. A separate interpolation identity writes the discrete energy as a continuous convolution term plus a positive lattice-roughness correction. Combining it with the sharp Euclidean Young inequality isolates the Gaussian constant

$$\Gamma = \frac{3\sqrt{3}}{4}.$$

These facts lead to the conjectural first-order asymptotic

$$t_n = 3 - (1 + o(1))\frac{\log \Gamma}{\log n}$$

for the optimal dimension-free additive-energy exponent on I_n^d. The present paper does not prove this asymptotic. Its missing input is the refinement-rigidity statement formulated in Conjecture 16.1: small defect on most active scales must produce compactness of a rescaled profile and hence a Gaussian continuum limit.

1.6. **The transverse epilogue.** Exact tensorization survives for every output exponent $r \geq 1$, but the binary boundary reduction is special to $r = 2$. For $r > 2$, the two orientations of two biased binary profiles are no longer degenerate, and the flat–biased equality point on the quadratic curved boundary has a strictly improving transverse direction. For each fixed $q \in (1, 4/3)$, Theorem 17.8 constructs and globalizes a real-analytic biased–biased equality branch for $2 \leq r < 2 + \varepsilon_q$. The first bias is linear in $r - 2$, whereas the correction to the sharp reciprocal exponent begins at order $(r - 2)^2$. Near the upper endpoint, a quartic normal form produces a supercritical pitchfork, and Theorem 17.21 closes the corresponding flat-to-biased transition globally. At $q = 1$ the exact threshold is $p = r$, but no uniform bridge from $q > 1$ to this endpoint and no single r-radius valid over the entire q-range are asserted.

1.7. **Proof architecture.** The proofs are organized around six structural reductions.

(1) *Product reduction.* On the binary cube, coordinate induction gives exact tensorization of the best constant. The d-dimensional problem is therefore exactly a two-vector problem, not merely bounded by one.
(2) *Hilbertian boundary geometry.* At $r = 2$, normalization by the two input ℓ^2 norms leaves an interaction depending only on the product of two balance parameters. In logarithmic coordinates every interior critical point has indefinite Hessian, so an extremum lies on a boundary where one input is flat or a point mass.
(3) *The ternary two-envelope mechanism.* A ternary profile is described by center mass and endpoint imbalance. Highly unbalanced endpoints are controlled by the sharp binary envelope; the complementary range is dominated coefficientwise by the reflection-symmetric ternary branch. The two estimates overlap and reduce the sharp problem to a one-variable phase portrait.
(4) *Variational reflection theory.* The cubic correlation field is the gradient of the quartic energy. Endpoint peeling closes in a finite Laurent-polynomial jet. For symmetric profiles, additive quadruples define a reversible Markov kernel; exchangeability and one nonnegative square give the exact odd spectral radius $1/3$.
(5) *Renormalization.* Parity splitting produces an exact scalar local gain and a nonnegative square defect. Iteration turns the global quotient into a pathwise multiplicative cascade. The one-scale atomic–smooth dichotomy identifies the only nearly lossless local configurations.
(6) *Continuum identification and transverse bifurcation.* Hat interpolation separates continuous Young deficit from lattice roughness. This identifies the Gaussian candidate for large alphabets. In the independent r-direction, implicit-function, compact-gap, and tail arguments globalize the symmetry-broken branches born at the Hilbertian point.

Numerical values are never used as definitions. The ternary constants are characterized analytically; the finite interval computations needed for strict overlap and uniqueness are collected in the appendix.

1.8. **Organization.** Sections 3–7 solve the binary quadratic problem and record its combinatorial, entropic, and Fourier consequences. Sections 8 and 9 treat the ternary threshold and complete norm curve. Sections 10–12 develop the cubic variational field, support rigidity, and reflection spectral theory for general alphabets. Sections 13–16 develop parity renormalization, the multiplicative cascade, the discrete–continuous bridge, and the remaining rigidity conjecture. Sections 17 and 18 form the transverse epilogue devoted to r near 2. The appendix contains the certified numerical enclosures used in the ternary overlap argument.

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2. Notation and elementary reductions

All discrete norms are taken with counting measure. A finite sequence is extended by zero outside its indicated support. Convolution on $\mathbb{Z}^d$ is

$$(f * g)(z) = \sum_{x+y=z} f(x)g(y),$$

and reflection is $\widetilde{f}(x) = f(-x)$. We use the usual continuous interpretations when an exponent equals ∞.

The coefficientwise inequality

$$|f * g| \le |f| * |g| \tag{2.1}$$

shows that every sharp norm problem considered below may first be reduced to nonnegative inputs. Complex phases re-enter only in the classification of equality. If f_i and g_i live on separate coordinates, then tensor products satisfy

$$\Big\| \bigotimes_i f_i * \bigotimes_i g_i \Big\|_r = \prod_i \|f_i * g_i\|_r, \qquad \Big\| \bigotimes_i f_i \Big\|_p = \prod_i \|f_i\|_p. \tag{2.2}$$

This gives the lower half of every exact tensorization statement; the upper half comes from slicing and Minkowski's inequality.

For self-convolution we write

$$Q(f) := \|f * f\|_2^2.$$

Translation and integer dilation preserve all additive relations and hence preserve the quotient $Q(f)/\|f\|_p^4$. This elementary invariance is used repeatedly when classifying supports. We distinguish carefully between the full bilinear problem, which is solved exactly on the binary alphabet, and the diagonal self-convolution problem, which is solved exactly on the ternary alphabet and studied structurally for larger alphabets.

3. Exact tensorisation and reduction to two points

For $p, q, r \in [1, \infty]$, define

$$\mathsf{C}_d(p, q; r) := \sup_{\substack{f,g \ne 0 \\ \operatorname{supp} f, \operatorname{supp} g \subseteq Q_d}} \frac{\|f * g\|_r}{\|f\|_p \|g\|_q}. \tag{3.1}$$

The following is a best-constant version of the tensorisation argument used in [4].

Proposition 3.1. *For every $d \geq 1$ and $p, q, r \in [1, \infty]$,*

$$\mathsf{C}_d(p, q; r) = \mathsf{C}_1(p, q; r)^d. \tag{3.2}$$

Moreover,

$$\mathsf{C}_1(p, q; r) = \sup_{x,y \geq 0} \frac{\left(1 + (x+y)^r + (xy)^r\right)^{1/r}}{(1 + x^p)^{1/p}(1 + y^q)^{1/q}}, \tag{3.3}$$

with the usual modifications when an exponent is infinity.

Proof. It is enough to consider nonnegative functions, since $|f * g| \leq |f| * |g|$. Split the last coordinate and write

$$f = (f_0, f_1), \qquad g = (g_0, g_1),$$

where the four slices are supported on Q_{d-1}. The three output slices are

$$f_0 * g_0, \qquad f_0 * g_1 + f_1 * g_0, \qquad f_1 * g_1.$$

For $r < \infty$, Minkowski's inequality and the definition of C_{d-1} give

$$\|f * g\|_r^r \leq \mathsf{C}_{d-1}(p, q; r)^r \sum_{k=0}^{2} \left(\sum_{i+j=k} \|f_i\|_p \|g_j\|_q \right)^r.$$

Applying the one-dimensional inequality to the two vectors

$$(\|f_0\|_p, \|f_1\|_p), \qquad (\|g_0\|_q, \|g_1\|_q)$$

shows that

$$\mathsf{C}_d(p, q; r) \leq \mathsf{C}_{d-1}(p, q; r)\mathsf{C}_1(p, q; r).$$

The same proof, with maxima in place of sums, applies when $r = \infty$. Iteration yields the upper bound in (3.2).

For the reverse inequality, take tensor powers of one-dimensional near-extremisers. Convolution and all three norms factor under tensor products, so the ratio is the dth power of the one-dimensional ratio. This proves (3.2).

Finally, a nonzero nonnegative function on Q_1 is a two-vector. By homogeneity, two such vectors can be written as scalar multiples of $(1, x)$ and $(1, y)$, with zero coordinates covered by limits. Their convolution is $(1, x + y, xy)$, which gives (3.3). □

For the remainder of the proof of Theorem 1.1, write

$$\mathsf{C}_d(p, q) := \mathsf{C}_d(p, q; 2).$$

By Proposition 3.1, it remains to determine $\mathsf{C}_1(p, q)$.

4. The Hilbertian two-point problem

Let $f = (a, b)$ and $g = (c, e)$ be nonnegative two-vectors. Direct expansion gives

$$\|f * g\|_2^2 = (a^2 + b^2)(c^2 + e^2) + 2abce. \tag{4.1}$$

Introduce the balance parameters

$$u := \frac{2ab}{a^2 + b^2}, \qquad v := \frac{2ce}{c^2 + e^2}. \tag{4.2}$$

They belong to $[0, 1]$, with $u = 0$ for a point mass and $u = 1$ for a constant vector. Equation (4.1) becomes

$$\frac{\|f * g\|_2}{\|f\|_2 \|g\|_2} = \left(1 + \frac{uv}{2}\right)^{1/2}. \tag{4.3}$$

For fixed u, the ratio $\|f\|_p / \|f\|_2$ is also determined by u. Parametrise

$$u = \operatorname{sech} z, \qquad z \geq 0, \tag{4.4}$$

and normalise $\|f\|_2 = 1$. Up to interchanging the two coordinates,

$$a^2 = \frac{e^z}{2\cosh z}, \qquad b^2 = \frac{e^{-z}}{2\cosh z}.$$

Thus

$$\Psi_p(u) := \frac{\|f\|_p}{\|f\|_2} = \frac{\big(2\cosh(pz/2)\big)^{1/p}}{(2\cosh z)^{1/2}}. \tag{4.5}$$

The main point of this section is that the maximum of the two-variable quotient occurs on an edge of the square $[0,1]^2$.

Proposition 4.1. *For all $p, q \in [1, \infty]$,*

$$\mathsf{C}_1(p,q)^2 = \max\left\{1,\ 2^{1-2/p} \max_{0\le t\le 1} \frac{1+t+t^2}{(1+t^q)^{2/q}},\right.$$
$$\left. 2^{1-2/q} \max_{0\le t\le 1} \frac{1+t+t^2}{(1+t^p)^{2/p}}\right\}. \tag{4.6}$$

In particular, apart from the point-mass alternative represented by the first term, one of the two inputs can be taken proportional to $(1,1)$*.*

Proof. We first assume $p, q < \infty$. Put

$$u = e^{-s}, \qquad v = e^{-t}, \qquad s, t \ge 0,$$

and define

$$A_p(s) := \log \Psi_p(e^{-s}), \qquad H(w) := \frac{1}{2}\log\left(1 + \frac{e^{-w}}{2}\right). \tag{4.7}$$

By (4.3), the logarithm of the reciprocal of the quotient to be maximised is

$$D(s,t) := A_p(s) + A_q(t) - H(s+t). \tag{4.8}$$

We show that D has no interior local minimum.

Write

$$s = \log\cosh z, \qquad \alpha := \frac{p}{2}.$$

Differentiating (4.5) gives

$$A_p'(s) = \frac{1}{2}\left(\frac{\tanh(\alpha z)}{\tanh z} - 1\right). \tag{4.9}$$

Suppose first that $1 \le p < 2$, so $0 < \alpha < 1$. Set

$$c_p(z) := \frac{\tanh(\alpha z)}{\tanh z}.$$

Then

$$A_p''(s) = \frac{c_p'(z)}{2\tanh z} > 0. \tag{4.10}$$

Indeed,

$$\frac{c_p'(z)}{c_p(z)} = \frac{2\alpha}{\sinh(2\alpha z)} - \frac{2}{\sinh(2z)} > 0,$$

because $x \mapsto \sinh x/x$ is strictly increasing on $(0,\infty)$.

We shall also need the upper bound

$$A_p''(s) < c_p(z)(1 - c_p(z)). \tag{4.11}$$

Using (4.10), this is equivalent to

$$\frac{\alpha}{\sinh(2\alpha z)} - \frac{1}{\sinh(2z)} < \tanh z - \tanh(\alpha z). \tag{4.12}$$

To verify it, define

$$G(r) := \tanh(rz) + \frac{r}{\sinh(2rz)}.$$

For $\alpha \le r \le 1$, writing $w = rz$ and using $z \ge w$, we obtain

$$\begin{aligned} G'(r) &= z \operatorname{sech}^2(rz) + \frac{1}{\sinh(2rz)} - \frac{2rz\cosh(2rz)}{\sinh^2(2rz)} \\ &\ge w \operatorname{sech}^2 w + \frac{1}{\sinh(2w)} - \frac{2w\cosh(2w)}{\sinh^2(2w)} \\ &= \frac{\sinh(2w) - 2w}{\sinh^2(2w)} > 0. \end{aligned}$$

Thus $G(1) > G(\alpha)$, which is exactly (4.12).

Assume now that $p, q < 2$ and that (s,t) is an interior critical point of D. Put

$$R := e^{-(s+t)}.$$

Since

$$H'(s+t) = -\frac{R}{2(2+R)}, \qquad H''(s+t) = \frac{R}{(2+R)^2}, \tag{4.13}$$

the critical-point equations and (4.9) imply

$$c_p(z) = c_q(\zeta) = \frac{2}{2+R}, \tag{4.14}$$

where $t = \log\cosh\zeta$. Consequently,

$$c_p(1-c_p) = c_q(1-c_q) = 2H''(s+t).$$

By (4.11),

$$0 < A_p''(s) < 2H''(s+t), \qquad 0 < A_q''(t) < 2H''(s+t). \tag{4.15}$$

The Hessian of D is

$$\begin{pmatrix} A_p'' - H'' & -H'' \\ -H'' & A_q'' - H'' \end{pmatrix}.$$

Writing $X = A_p''/H''$ and $Y = A_q''/H''$, its determinant divided by $(H'')^2$ is

$$XY - X - Y = (X-1)(Y-1) - 1 < 0$$

by (4.15). Thus every interior critical point is a saddle.

When $p, q < 2$, the functions A_p, A_q are nonnegative and tend to zero at infinity. Hence

$$\liminf_{s+t\to\infty} D(s,t) \ge 0.$$

If the infimum of D were smaller than the minimum of its two finite boundary infima and zero, that lower value would be attained at an interior local minimum, a contradiction. Therefore the infimum occurs on $s = 0$, on $t = 0$, or at infinity.

If $p \ge 2$, then (4.9) gives $A_p'(s) \ge 0$, whereas $H' < 0$. Thus $\partial_s D > 0$, and the minimum in the s variable is at $s = 0$. The same argument applies to $q \ge 2$. We have therefore proved the boundary reduction for all finite p, q. The cases involving infinity follow by passage to the limit.

It remains to compute the two finite edges. On $s = 0$, the first vector is proportional to $(1,1)$. After scaling and interchanging the coordinates of the other vector, write it as $(1,t)$ with $0 \le t \le 1$. Then

$$\|(1,1) * (1,t)\|_2^2 = 2(1 + t + t^2),$$

and

$$\|(1,1)\|_p^2 \|(1,t)\|_q^2 = 2^{2/p}(1+t^q)^{2/q}.$$

This gives the second term in (4.6); the other edge is symmetric. At infinity one input is a point mass, giving the term 1. □

5. The one-variable phase diagram

We now analyse the maximum in (1.4).

Lemma 5.1. *The function* Λ *has the following form.*

(i) $\Lambda(1) = 0$.

(ii) *If* $q \geq 4/3$*, then*

$$\Lambda(q) = \log_2 3 - \frac{2}{q},$$

and the maximum is attained at $t = 1$.

(iii) *If* $1 < q < 4/3$*, then the maximum is attained at a unique* $t_q \in (0,1)$*, determined by*

$$t_q^{q-1}(2 + t_q) = 1 + 2t_q.$$

Proof. The case $q = \infty$ is immediate from the definition. For $q < \infty$, set

$$F_q(t) := \log(1 + t + t^2) - \frac{2}{q}\log(1 + t^q), \qquad 0 \leq t \leq 1.$$

A direct calculation gives

$$F_q'(t) = \frac{1 + 2t - t^{q-1}(2 + t)}{(1 + t + t^2)(1 + t^q)}. \tag{5.1}$$

For $q = 1$, one has $F_1(t) \leq 0 = F_1(0)$, proving (i).

Suppose $q \geq 4/3$. Since $0 \leq t \leq 1$,

$$t^{q-1} \leq t^{1/3}.$$

Writing $u = t^{1/3}$,

$$1 + 2t - t^{1/3}(2 + t) = 1 + 2u^3 - 2u - u^4 = (1 - u)^3(1 + u) \geq 0.$$

Thus F_q is increasing and its maximum is $F_q(1) = \log 3 - (2/q)\log 2$. This proves (ii).

Now let $1 < q < 4/3$ and put

$$K_q(t) := t^{q-1}(2 + t) - (1 + 2t).$$

Then

$$K_q''(t) = (q - 1)t^{q-3}\big(2(q - 2) + qt\big) < 0 \tag{5.2}$$

for $0 < t < 1$. Also,

$$K_q(0+) = -1, \qquad K_q(1) = 0, \qquad K_q'(1) = 3q - 4 < 0.$$

Strict concavity therefore gives a unique zero $t_q \in (0,1)$. By (5.1), F_q increases before t_q and decreases afterwards. This proves (iii). ☐

The two inequalities in (1.8) simplify in either half of the phase diagram.

Lemma 5.2. *The function*

$$q \longmapsto \Lambda(q) + \frac{2}{q}$$

is nonincreasing on $[1, \infty]$.

Proof. For fixed $t \in [0,1]$, consider

$$\log_2(1 + t + t^2) + \frac{2}{q}\log_2\frac{2}{1 + t^q}.$$

It is enough to show that the second term is nonincreasing in q. For $0 < t < 1$, write $a = -\log t$ and

$$h(x) := \log\frac{2}{1 + e^{-x}}.$$

Then h is concave, increasing, and $h(0)=0$, while

$$\frac{1}{q}\log\frac{2}{1+t^q}=a\frac{h(aq)}{aq}.$$

The quotient $h(x)/x$ is nonincreasing for a concave function vanishing at zero. The endpoint cases $t=0,1$ are immediate. Taking the supremum over t proves the claim. □

Proof of Theorem 1.1 and Corollary 1.2. Combining Proposition 4.1 with the definition of Λ gives

$$\mathsf{C}_1(p,q)=2^{\frac{1}{2}\mathsf{M}(p,q)}. \tag{5.3}$$

The exact tensorisation (3.2) now proves (1.6) and its sharpness.

Since point masses show that the best constant is always at least one, the constant-one inequality holds exactly when $\mathsf{M}(p,q)=0$, which is equivalent to (1.8). If $p\geq q$, then Lemma 5.2 gives

$$\Lambda(q)+\frac{2}{q}\geq\Lambda(p)+\frac{2}{p}.$$

Equivalently,

$$1-\frac{2}{p}+\Lambda(q)\geq 1-\frac{2}{q}+\Lambda(p),$$

so only the first nontrivial term in (1.5) needs to be checked. This proves (1.9). □

For completeness, the curved part of the boundary admits a direct parametrisation. Solving (1.11) for q gives

$$q(t)=1+\frac{\log\frac{1+2t}{2+t}}{\log t},\qquad 0<t<1. \tag{5.4}$$

The corresponding boundary point is

$$\frac{1}{p(t)}=\frac{1}{2}+\frac{1}{2}\left[\log_2(1+t+t^2)-\frac{2}{q(t)}\log_2(1+t^{q(t)})\right],\qquad \frac{1}{q}=\frac{1}{q(t)}. \tag{5.5}$$

As $t\downarrow 0$, the point tends to $(1/2,1)$; as $t\uparrow 1$, it tends to

$$\left(\frac{\log_2 3}{2}-\frac{1}{4},\frac{3}{4}\right).$$

On the curved boundary, equality in the one-dimensional inequality is attained by a constant vector and a biased vector $(1,t_q)$. Their tensor powers give equality in every dimension. On the linear part $p,q\geq 4/3$, both inputs may be taken constant. At the endpoint $(p,q)=(2,1)$ one input is constant and the other is a point mass.

6. Cross-additive energy and sumsets

For nonempty sets $A,B\subseteq Q_d$, define

$$E_+(A,B):=\sum_{x\in\mathbb{Z}^d}(\mathbf{1}_A*\mathbf{1}_B)(x)^2. \tag{6.1}$$

Equivalently, $E_+(A,B)$ counts solutions to

$$a_1+b_1=a_2+b_2,\qquad a_1,a_2\in A,\quad b_1,b_2\in B.$$

The usual difference energy is covered as well. If $\mathbf{1}=(1,\dots,1)$ and $\widetilde{B}:=\{\mathbf{1}-b:b\in B\}$, then

$$a_1+(\mathbf{1}-b_1)=a_2+(\mathbf{1}-b_2)\quad\Longleftrightarrow\quad a_1-b_1=a_2-b_2,$$

and $|\widetilde{B}|=|B|$.

Let

$$\mathsf{K}_d(p,q):=\sup_{\varnothing\neq A,B\subseteq Q_d}\frac{E_+(A,B)^{1/2}}{|A|^{1/p}|B|^{1/q}}. \tag{6.2}$$

The functional and set constants need not agree in a fixed dimension, because the one-dimensional extremiser on the curved boundary is weighted. They nevertheless have the same exact exponential rate. The diagonal estimate $E_+(A,A) \le |A|^{\log_2 6}$ goes back to Kane and Tao [10]; higher-energy extensions and related cube inequalities appear in [8, 4].

Theorem 6.1. *For all $d \ge 1$ and $p, q \in [1,\infty]$,*

$$\frac{\mathsf{C}_1(p,q)^d}{d+1} \le \mathsf{K}_d(p,q) \le \mathsf{C}_1(p,q)^d. \tag{6.3}$$

Consequently,

$$\lim_{d\to\infty} \mathsf{K}_d(p,q)^{1/d} = \mathsf{C}_1(p,q) = 2^{\mathsf{M}(p,q)/2}. \tag{6.4}$$

In particular,

$$E_+(A,B) \le |A|^{2/p}|B|^{2/q} \tag{6.5}$$

holds for every d and all $A, B \subseteq Q_d$ if and only if $\mathsf{M}(p,q) = 0$.

Proof. The upper bound in (6.3) is Theorem 1.1 applied to indicator functions.

For the lower bound, the case $\mathsf{C}_1(p,q) = 1$ follows from singletons. Suppose $\mathsf{C}_1(p,q) > 1$. Since $\mathsf{K}_d(p,q) = \mathsf{K}_d(q,p)$, Proposition 4.1 allows us, after possibly interchanging p and q, to choose $t \in [0,1]$ such that the one-dimensional extremal pair can be taken as

$$f_1 = (1,1), \qquad g_1 = (1,t).$$

Let

$$F = f_1^{\otimes d} = \mathbf{1}_{Q_d}, \qquad G = g_1^{\otimes d}.$$

Then $G(x) = t^{|x|}$, where $|x|$ is the Hamming weight, and

$$\|F * G\|_2 = \mathsf{C}_1(p,q)^d \|F\|_p \|G\|_q. \tag{6.6}$$

Decompose G into Hamming layers:

$$G = \sum_{m=0}^{d} t^m \mathbf{1}_{S_m}, \qquad S_m := \{x \in Q_d : |x| = m\}.$$

By the triangle inequality and the definition of $\mathsf{K}_d(p,q)$,

$$\begin{aligned} \|F * G\|_2 &\le \sum_{m=0}^{d} t^m \|\mathbf{1}_{Q_d} * \mathbf{1}_{S_m}\|_2 \\ &\le \mathsf{K}_d(p,q)\|F\|_p \sum_{m=0}^{d} t^m |S_m|^{1/q}. \end{aligned}$$

For $q < \infty$, Hölder's inequality in the index m gives

$$\sum_{m=0}^{d} t^m |S_m|^{1/q} \le (d+1)^{1-1/q} \left(\sum_{m=0}^{d} t^{mq} |S_m| \right)^{1/q} \le (d+1)\|G\|_q.$$

The same final bound is immediate for $q = \infty$. Comparing with (6.6) proves the lower bound in (6.3). The limit follows.

If $\mathsf{M}(p,q) = 0$, (6.5) follows from Theorem 1.1. If $\mathsf{M}(p,q) > 0$, then (6.4) shows that $\mathsf{K}_d(p,q) > 1$ for all sufficiently large d, so (6.5) cannot hold in every dimension. ☐

The sumset consequence is immediate but useful in asymmetric situations.

Corollary 6.2. *Let $A, B \subseteq Q_d$ be nonempty. For every (p,q) satisfying (1.8),*

$$|A+B| \geq |A|^{2-2/p}|B|^{2-2/q}. \tag{6.7}$$

Equivalently,

$$|A+B| \geq \sup_{\mathsf{M}(p,q)=0} |A|^{2-2/p}|B|^{2-2/q}. \tag{6.8}$$

Proof. Writing $r_{A+B} = \mathbf{1}_A * \mathbf{1}_B$, Cauchy–Schwarz gives

$$|A|^2|B|^2 = \left(\sum_x r_{A+B}(x)\right)^2 \leq |A+B| \sum_x r_{A+B}(x)^2.$$

Apply (6.5) and rearrange. □

Remark 6.3. For $A = B$, the diagonal point recovers the usual energy-based estimate

$$|A+A| \geq |A|^{4-\log_2 6}.$$

The sharp binary Brunn–Minkowski estimate

$$|A+B| \geq (|A||B|)^{\log_2 3/2}$$

is stronger in the balanced case and comes from a reverse/sup-convolution inequality rather than from L^2 multiplicity control. The advantage of (6.8) is its asymmetric dependence on $|A|$ and $|B|$.

7. Entropy and Fourier formulations

For a finitely supported probability mass function P and $s \in (0,\infty) \setminus \{1\}$, write

$$H_s(P) := \frac{1}{1-s} \log_2 \sum_x P(x)^s.$$

We use the standard continuous extensions at $s = 1$ and $s = \infty$. If X is a random variable with law P_X, write $H_s(X) := H_s(P_X)$.

Corollary 7.1. *Let X, Y be independent random variables taking values in Q_d. Then, for all $p, q \in [1,\infty]$,*

$$H_2(X+Y) \geq 2\left(1 - \frac{1}{p}\right) H_p(X) + 2\left(1 - \frac{1}{q}\right) H_q(Y) - d\mathsf{M}(p,q). \tag{7.1}$$

The dimension penalty $d\mathsf{M}(p,q)$ is optimal. In the region (1.8), it vanishes.

Proof. Apply Theorem 1.1 to P_X and P_Y. Since

$$\log_2 \|P\|_s = -\left(1 - \frac{1}{s}\right) H_s(P),$$

taking logarithms of (1.6) gives (7.1). Conversely, every nonnegative pair of functions can be normalised to probability mass functions without changing the ratio in (1.6). Thus optimality is equivalent to the optimality in Theorem 1.1. □

Let

$$\widehat{f}(\xi) := \sum_{x\in\mathbb{Z}^d} f(x) e^{-2\pi i x\cdot\xi}, \qquad \xi \in \mathbb{T}^d.$$

Plancherel gives the following exact reformulation.

Corollary 7.2. *For $p, q \in [1,\infty]$ and f, g supported on Q_d,*

$$\left(\int_{\mathbb{T}^d} |\widehat{f}(\xi)|^2 |\widehat{g}(\xi)|^2 \, \mathrm{d}\xi\right)^{1/2} \leq 2^{\frac{d}{2}\mathsf{M}(p,q)} \|f\|_p \|g\|_q. \tag{7.2}$$

The constant is optimal.

Proof. Since $\widehat{f * g} = \widehat{f}\,\widehat{g}$,

$$\|f * g\|_2^2 = \int_{\mathbb{T}^d} |\widehat{f}|^2 |\widehat{g}|^2.$$

The result is therefore identical to Theorem 1.1. $\square$

8. The ternary threshold and the two-envelope mechanism

We first give a self-contained account of the exact ternary calculation. The classical Hausdorff–Young inequality gives $\|\widehat{f}\|_{L^4} \le \|f\|_{\ell^{4/3}}$; the sharp Euclidean forms of Hausdorff–Young and Young's convolution inequality go back to Beckner and Brascamp–Lieb [1, 6]. For the binary cube, Kane and Tao proved the sharp additive-energy estimate $E(A) \le |A|^{\log_2 6}$ [10]. Larger alphabets were studied in [8, 14, 3].

The dimension-compression theorem of Beker, Crmarić, and Kovač implies that the ternary constant-one problem is equivalent to a single-coordinate three-variable inequality.

Proposition 8.1 (Beker–Crmarić–Kovač)**.** *Let $p, t > 0$ satisfy $pt = 4$. The following are equivalent.*

- (i) *Every nonnegative $f\colon \mathbb{Z} \to \mathbb{R}$ supported in $\{0, 1, 2\}$ satisfies $\|f\|_{\mathrm{U}^2} \le \|f\|_{\ell^p}$.*
- (ii) *The same inequality holds in every dimension for every complex-valued function supported in $\{0, 1, 2\}^d$.*
- (iii) *Every $A \subseteq \{0, 1, 2\}^d$ satisfies $E(A) \le |A|^t$ in every dimension.*
- (iv) *There is a dimension-independent constant in the corresponding additive-energy bound with exponent t.*

For $f = (a, b, c)$ with $a, b, c \ge 0$,

$$f * f = (a^2, 2ab, b^2 + 2ac, 2bc, c^2),$$

so

$$Q(a, b, c) = a^4 + b^4 + c^4 + 4a^2b^2 + 4b^2c^2 + 4a^2c^2 + 4ab^2c. \tag{8.1}$$

Equivalently, with

$$C_0 = a^2 + b^2 + c^2, \qquad C_1 = ab + bc, \qquad C_2 = ac,$$

one has

$$Q(a, b, c) = C_0^2 + 2C_1^2 + 2C_2^2. \tag{8.2}$$

The two terms C_1, C_2 are the two nontrivial autocorrelation lags.

8.1. **The symmetric critical branch.**

Write $p = 2q$ and $t = 2/q = 4/p$. For the symmetric profile $(1, \sqrt{x}, 1)$, define

$$R_q(x) := \frac{x^2 + 12x + 6}{(x^q + 2)^{2/q}}, \qquad x \ge 0. \tag{8.3}$$

For $x > 1$, put

$$s(x) := 1 - \frac{\log(3(x+1)/(x+6))}{\log x}, \tag{8.4}$$

$$e(x) := 2 - \frac{2\log(3(x+1))}{\log(x^2 + 12x + 6)}, \tag{8.5}$$

$$\Psi(x) := s(x) - e(x). \tag{8.6}$$

Lemma 8.2. *The function Ψ has a unique zero x_* in $(3, 31/10)$. If*

$$q_* := s(x_*), \qquad p_* := 2q_*, \qquad t_* := \frac{2}{q_*}, \tag{8.7}$$

then

$$x_*^{1-q_*}(x_* + 6) = 3(x_* + 1), \qquad R_{q_*}(x_*) = 1. \tag{8.8}$$

Moreover,

$$1.467 < p_* < 1.477, \qquad t_* > \log_2 6. \tag{8.9}$$

Proof. Differentiating (8.4)–(8.5), with $M = x^2 + 12x + 6$, gives

$$s'(x) = -\frac{5}{(x+1)(x+6)\log x} + \frac{\log(3(x+1)/(x+6))}{x(\log x)^2},$$
$$e'(x) = -\frac{2}{(x+1)\log M} + \frac{4(x+6)\log(3(x+1))}{M(\log M)^2}.$$

The elementary interval bounds recorded in Section A imply $\Psi'(x) < -0.0105$ on $[3, 3.1]$, while

$$\Psi(3) > 0.00213, \qquad \Psi(3.1) < -0.00099.$$

Hence there is a unique zero x_*. The identity $q_* = s(x_*)$ is the first equation in (8.8); it also yields

$$x_*^{q_*} + 2 = \frac{x_*^2 + 12x_* + 6}{3(x_*+1)}. \tag{8.10}$$

The equation $s(x_*) = e(x_*)$, together with (8.10), gives $R_{q_*}(x_*) = 1$. The numerical inequalities in (8.9) follow from the same certified interval calculation. □

Lemma 8.3. *For every $x \geq 0$,*

$$R_{q_*}(x) \leq 1. \tag{8.11}$$

Equivalently, for every $u \in [0, 1]$,

$$u^{t_*} + \frac{12}{2^{t_*/2}}[u(1-u)]^{t_*/2} + \frac{6}{2^{t_*}}(1-u)^{t_*} \leq 1. \tag{8.12}$$

Proof. The sign of $R'_q(x)$ is the sign of $G_q(x) - 3$, where

$$G_q(x) := x^{1-q}\frac{x+6}{x+1}. \tag{8.13}$$

Furthermore,

$$\frac{G'_q(x)}{G_q(x)} = \frac{1-q}{x} - \frac{5}{(x+6)(x+1)}, \tag{8.14}$$

whose sign is controlled by the quadratic

$$P_q(x) := (1-q)(x+6)(x+1) - 5x.$$

At $q = q_*$ the equation $G_{q_*}(x) = 3$ has exactly three positive roots, with x_* the middle root. Consequently R_{q_*} has a unique finite local maximum at x_*. Since

$$R_{q_*}(x_*) = 1, \qquad R_{q_*}(0) = \frac{6}{2^{t_*}} < 1, \qquad \lim_{x\to\infty} R_{q_*}(x) = 1,$$

we obtain (8.11). The substitution $u/(1-u) = x^{q_*}/2$ gives (8.12). □

8.2. **The binary envelope and coefficient geometry.**

Lemma 8.4. *For every $t \geq 2$ and $u \in [0, 1]$,*

$$u^t + (1-u)^t + (2^t - 2)[u(1-u)]^{t/2} \leq 1. \tag{8.15}$$

The coefficient $2^t - 2$ is optimal.

Proof. Optimality follows at $u = 1/2$. Put $u = v/(1+v)$ and reduce by symmetry to $0 \leq v \leq 1$. Writing $v = e^{-2r}$, inequality (8.15) becomes

$$(2\cosh r)^t - 2\cosh(tr) \geq 2^t - 2.$$

After differentiating, it is enough to prove

$$\frac{1-x^t}{1-x} \leq (1+x)^{t-1}, \qquad 0 \leq x \leq 1.$$

With $y = (1-x)/(1+x)$ this is

$$(1+y)^t - (1-y)^t \le 2^t y.$$

The left side is convex, vanishes at 0, and equals 2^t at 1, so it lies below its endpoint chord. $\square$

For an arbitrary profile, assume by reflection that $a \ge c$, and set

$$\theta := \frac{c}{a} \in [0,1], \qquad u := \frac{b^p}{a^p + b^p + c^p}. \tag{8.16}$$

Define

$$\alpha_p(\theta) := \frac{4(1+\theta+\theta^2)}{(1+\theta^p)^{2/p}}, \qquad \beta_p(\theta) := \frac{1+4\theta^2+\theta^4}{(1+\theta^p)^{4/p}}. \tag{8.17}$$

Then the exact identity

$$\frac{Q(a,b,c)}{(a^p+b^p+c^p)^{4/p}} = u^t + \alpha_p(\theta)[u(1-u)]^{t/2} + \beta_p(\theta)(1-u)^t, \qquad t = \frac{4}{p}, \tag{8.18}$$

holds.

Lemma 8.5. *Let $4/3 \le p < 2$. The function α_p is increasing on $[0,1]$. The function β_p has no interior local maximum: it first decreases and then increases for $p > 4/3$, and is decreasing for $p = 4/3$.*

Proof. For $0 < \theta < 1$, the sign of $(\log \alpha_p)'$ is the sign of

$$1 + 2\theta - \theta^{p-1}(2+\theta),$$

which is nonnegative because

$$\theta^{p-1}(2+\theta) \le \theta^{1/3}(2+\theta) \le 1 + 2\theta.$$

For β_p, after differentiating and clearing positive factors, the sign is that of

$$h(s) := s^\delta(2+s) - (1+2s), \qquad s = \theta^2, \qquad \delta = \frac{2-p}{2}.$$

For $p > 4/3$, strict concavity of h on $(0,1)$ and its endpoint behavior show that it has exactly one zero; for $p = 4/3$ one has $h(r^3) = (r-1)^3(r+1) \le 0$. $\square$

Lemma 8.6. *At $p = p_*$,*

$$\alpha_{p_*}(1/3) < 2^{t_*} - 2, \tag{8.19}$$

and

$$\beta_{p_*}(1/3) < \beta_{p_*}(1) = \frac{6}{2^{t_*}}. \tag{8.20}$$

Proof. The first inequality follows from $p_* < 1.477$ and the certified bounds

$$\alpha_{1.477}(1/3) < 4.528 < 4.535 < 2^{4/1.477} - 2.$$

For the second, one has

$$\frac{\beta_p(1/3)}{\beta_p(1)} = \frac{59}{243}\left(\frac{2}{1+3^{-p}}\right)^{4/p}.$$

The last factor is strictly decreasing in p, and evaluation at $p = 1.467 < p_*$ gives a value below 0.979. $\square$

8.3. **Completion of the sharp threshold.**

Proof of Theorem 1.4. It is enough, by Proposition 8.1, to prove the one-dimensional inequality at $p = p_*$. If $0 \le \theta \le 1/3$, then

$$\alpha_{p_*}(\theta) < 2^{t_*} - 2, \qquad \beta_{p_*}(\theta) \le 1,$$

and Lemma 8.4 controls (8.18). If $1/3 \le \theta \le 1$, then

$$\alpha_{p_*}(\theta) \le \alpha_{p_*}(1), \qquad \beta_{p_*}(\theta) \le \beta_{p_*}(1),$$

by Lemmas 8.5 and 8.6. The resulting expression is bounded by the centered envelope (8.12). Equality occurs for the symmetric profile $(1, \sqrt{x_*}, 1)$ and for point masses. Monotonicity of finite-dimensional ℓ^p quasi-norms gives all $p < p_*$. The profile $(1, \sqrt{x_*}, 1)$ rules out every $p > p_*$. The additive-energy statement then follows from Proposition 8.1. □

9. The complete ternary best-constant curve

We now upgrade the constant-one threshold to the exact one-dimensional operator norm for every $p > 0$.

9.1. **A target-constant envelope.**

Lemma 9.1. *Let $t \ge 2$ and $M \ge 1$. Then for every $u \in [0, 1]$,*

$$u^t + (1-u)^t + (2^t M - 2)[u(1-u)]^{t/2} \le M. \tag{9.1}$$

Proof. Let $w = [u(1-u)]^{t/2} \le 2^{-t}$. By Lemma 8.4,

$$\begin{aligned} &u^t + (1-u)^t + (2^t M - 2)w \\ &\quad = \left[u^t + (1-u)^t + (2^t - 2)w\right] + 2^t(M-1)w \\ &\quad \le 1 + (M-1) = M. \end{aligned}$$

□

Set

$$p_{\text{bin}} := \frac{4}{\log_2 6} = 1.5474112289\ldots. \tag{9.2}$$

For $p \ge p_*$ define

$$M_p := \sup_{0 \le u \le 1} \left\{u^t + \alpha_p(1)[u(1-u)]^{t/2} + \beta_p(1)(1-u)^t\right\}, \qquad t = \frac{4}{p}. \tag{9.3}$$

This is the sharp quotient restricted to reflection-symmetric profiles.

Lemma 9.2. *For every $p \in [p_*, p_{\text{bin}}]$,*

$$\alpha_p(1/3) \le \alpha_p(1) + \beta_p(1) - 1, \tag{9.4}$$

and for every $p \ge p_*$,

$$\beta_p(1/3) \le \beta_p(1). \tag{9.5}$$

Proof. The ratio in the proof of Lemma 8.6 is decreasing in p, so (9.5) follows from its strict inequality at p_*.

For (9.4), put

$$\Delta(p) := \alpha_p(1) + \beta_p(1) - 1 - \alpha_p(1/3).$$

Since $p_* > 1.467$ and $p_* < 1.477$, monotonicity in p and the certified bounds in Section A give

$$\Delta(p_*) > 4.5706 - 4.5273 > 0.0433.$$

Let $r = 3^{-p}$. Direct differentiation yields

$$\begin{aligned} \frac{p^2}{2}\Delta'(p) &= (\log 2)\alpha_p(1) + 2(\log 2)\beta_p(1) \\ &\quad - \alpha_p(1/3)\left(\log(1+r) + \frac{p(\log 3)r}{1+r}\right). \end{aligned} \tag{9.6}$$

On $[p_*, p_{\text{bin}}]$ one has

$$r < \frac{1}{5}, \quad p < 1.55, \quad \alpha_p(1) > 4.2, \quad \beta_p(1) > 0.75, \quad \alpha_p(1/3) < \frac{52}{9}.$$

Using $\log 2 > 0.69$ and $\log 3 < 1.1$, the right side of (9.6) is larger than

$$0.69(4.2 + 1.5) - \frac{52}{9}\big(0.2 + 1.55 \cdot 1.1 \cdot 0.2\big) > 0.$$

Thus Δ is increasing and remains positive throughout the interval. $\square$

9.2. **Global reduction to the symmetric branch.**

Proposition 9.3 (Global reduction to symmetric profiles)**.** *For every $p \geq p_*$ and every $a, b, c \geq 0$,*

$$\frac{Q(a,b,c)}{(a^p + b^p + c^p)^{4/p}} \leq M_p. \tag{9.7}$$

Proof. Use the variables θ, u in (8.16).

First suppose $p_* \leq p \leq p_{\text{bin}}$. Then $t \geq \log_2 6$, so the binary constant-one inequality gives

$$\beta_p(\theta) \leq 1 \qquad (0 \leq \theta \leq 1).$$

If $0 \leq \theta \leq 1/3$, then

$$\alpha_p(\theta) \leq \alpha_p(1/3).$$

Since M_p dominates its value at $u = 1/2$,

$$2^t M_p - 2 \geq \alpha_p(1) + \beta_p(1) - 1 \geq \alpha_p(1/3).$$

The target envelope Lemma 9.1 now proves (9.7).

If $1/3 \leq \theta \leq 1$, then

$$\alpha_p(\theta) \leq \alpha_p(1), \qquad \beta_p(\theta) \leq \beta_p(1)$$

by Lemmas 8.5 and 9.2. Hence the quotient is bounded coefficientwise by the symmetric expression defining M_p.

Next suppose $p_{\text{bin}} \leq p < 2$. Here $\beta_p(1) \geq \beta_p(0) = 1$. Since β_p has no interior local maximum,

$$\beta_p(\theta) \leq \beta_p(1),$$

and the same coefficientwise comparison applies.

Finally let $p \geq 2$. The proof of monotonicity for α_p remains valid. For β_p, the derivative has the sign of

$$h(s) = s^{\delta}(2+s) - (1+2s), \qquad \delta = \frac{2-p}{2} \leq 0.$$

Since $s^{\delta} \geq 1$ on $(0,1]$,

$$h(s) \geq (2+s) - (1+2s) = 1 - s \geq 0.$$

Thus both coefficients are increasing, and the symmetric profile again dominates coefficientwise. $\square$

9.3. **The symmetric phase portrait.** We next solve the symmetric one-variable problem completely.

Lemma 9.4. *Let $q > q_*$ and set $p = 2q$. The function R_q has a unique global maximizer $x_q \in (1, x_*)$, characterized by*

$$x_q^{1-q}(x_q + 6) = 3(x_q + 1). \tag{9.8}$$

The map $q \mapsto x_q$ is strictly decreasing,

$$x_{q_*} = x_*, \qquad x_1 = \frac{3}{2}, \qquad \lim_{q\to\infty} x_q = 1.$$

Proof. The sign of R_q' is the sign of $G_q - 3$, with G_q as in (8.13).

If $q \geq 1$, then (8.14) is strictly negative, so G_q is strictly decreasing. It crosses 3 exactly once, at a point larger than 1 because $G_q(1) = 7/2$. This crossing is the unique global maximum of R_q.

Suppose $q_* < q < 1$. The quadratic P_q controlling G_q' satisfies $P_q(0) > 0$, $P_q(1) = 9 - 14q < 0$, and $P_q(x) \to \infty$. Thus G_q has one local maximum and one local minimum. Furthermore,

$$G_q(0) = 0, \qquad G_q(1) = \frac{7}{2} > 3, \qquad G_q(x_*) < G_{q_*}(x_*) = 3, \qquad G_q(x) \to \infty.$$

Hence $G_q = 3$ has exactly three roots, and its middle root x_q lies in $(1, x_*)$. It is the unique finite local maximum of R_q.

To see that this maximum is global, note first that

$$R_q(x_*) > R_{q_*}(x_*) = 1,$$

because for a fixed nonzero vector its ℓ^q norm decreases as q increases. Also

$$\frac{R_q(1)}{R_q(0)} = \frac{19}{6}\left(\frac{2}{3}\right)^{2/q} > 1.$$

Indeed, the ratio is increasing in q, and at $q = q_*$ the inequality follows from $t_* = 2/q_* < 4/1.467 < \log_{3/2}(19/6)$. Since $R_q(x_q) > R_q(1)$ and $\lim_{x\to\infty} R_q(x) = 1$, the point x_q is the global maximizer.

For fixed $x > 1$, $G_q(x)$ is strictly decreasing in q. At the relevant crossing G_q decreases in x, so the implicit root x_q decreases in q. The remaining identities follow directly from (9.8). □

Proof of Theorem 1.5. For $p \leq p_*$, Theorem 1.4 gives $\mathcal{C}_3(p) \leq 1$, while point masses give equality. The right side of (1.14) also has supremum 1.

For $p > p_*$, Proposition 9.3 reduces the global quotient to symmetric profiles, and Lemma 9.4 gives the unique maximizing parameter $x_p = x_{p/2}$. At a critical point, (1.15) implies

$$x_p^{p/2} + 2 = \frac{x_p^2 + 12x_p + 6}{3(x_p + 1)}. \tag{9.9}$$

Substitution into $R_{p/2}(x_p)$ gives (1.16).

At $p = 2$, (1.15) gives $x_p = 3/2$, and direct substitution yields $15/7$. At $p = 4$ one obtains $x_p = (-1 + \sqrt{19})/3$ and $\mathcal{C}_3(4)^4 = 2 + \sqrt{19}$. As $p \to \infty$, $x_p \to 1$ and the quotient tends to 19.

The strict coefficient comparisons in the proof of Proposition 9.3, together with uniqueness in Lemma 9.4, give the stated uniqueness of the nonnegative maximizer for $p > p_*$. □

9.4. **Tensorization.** The one-coordinate constant tensorizes through the range relevant to the L^4 problem.

Proposition 9.5 (Exact tensorization for $p \le 4$)**.** *For $0 < p \le 4$, let*

$$\mathcal{C}_{3,d}(p) := \sup_{0 \neq f,\, \operatorname{supp} f \subseteq \{0,1,2\}^d} \frac{\|\widehat{f}\|_{L^4(\mathbb{T}^d)}}{\|f\|_{\ell^p}}.$$

Then

$$\mathcal{C}_{3,d}(p) = \mathcal{C}_3(p)^d. \tag{9.10}$$

Proof. The lower bound follows from tensor powers of a one-dimensional extremizer. For the upper bound, write f_j for the slices in the final coordinate. Applying the one-dimensional inequality pointwise in the remaining Fourier variables gives

$$\|\widehat{f}(\xi', \cdot)\|_{L^4} \le \mathcal{C}_3(p) \left(\sum_j |\widehat{f}_j(\xi')|^p \right)^{1/p}.$$

Since $4/p \ge 1$, Minkowski's inequality yields

$$\left\| \left(\sum_j |\widehat{f}_j|^p \right)^{1/p} \right\|_{L^4} \le \left(\sum_j \|\widehat{f}_j\|_{L^4}^p \right)^{1/p}.$$

Induction on d proves the upper bound. $\square$

10. The cubic correlation field and endpoint algebra

For a finite real sequence f, extended by zero outside its support, define

$$\mathcal{C}_f(k) := (f * f * \widetilde{f})(k), \qquad \widetilde{f}(j) := f(-j). \tag{10.1}$$

This is exactly the gradient of the quartic energy.

Lemma 10.1. *For every index k,*

$$\frac{\partial Q}{\partial f_k} = 4\mathcal{C}_f(k). \tag{10.2}$$

Consequently, a positive critical point of $Q(f)/\|f\|_p^4$ satisfies

$$\mathcal{C}_f(k) = \lambda f_k^{p-1} \tag{10.3}$$

for a constant $\lambda > 0$.

Proof. Differentiate $Q(f) = \sum_m (f * f)(m)^2$ and use commutativity of convolution. The Euler–Lagrange equation follows by a Lagrange multiplier under the normalization $\sum_k f_k^p = 1$. $\square$

Let $N = n - 1$ and peel off the endpoints:

$$f = a\delta_0 + g + c\delta_N, \qquad \operatorname{supp} g \subseteq \{1, \dots, N-1\}. \tag{10.4}$$

Put

$$M_2 := \|g\|_2^2, \qquad D := (g * g)(N), \qquad T_L := \mathcal{C}_g(0), \qquad T_R := \mathcal{C}_g(N). \tag{10.5}$$

Proposition 10.2 (Exact endpoint expansion)**.** *With the notation above,*

$$\begin{aligned} Q(f) = Q(g) + a^4 + c^4 + 4(a^2 + c^2)M_2 + 4a^2c^2 \\ + 4acD + 4aT_L + 4cT_R. \end{aligned} \tag{10.6}$$

In particular,

$$Q(a, g, c) - Q(c, g, a) = 4(a - c)(T_L - T_R). \tag{10.7}$$

Proof. Expand

$$f * f = g * g + 2(a\delta_0 + c\delta_N) * g + (a\delta_0 + c\delta_N)^2$$

and use the disjointness of the relevant shifted interior supports. The terms $\langle g * g, g\rangle$ and $\langle g * g, \tau_N g\rangle$ are precisely T_L and T_R, while $\langle g * g, \delta_N\rangle = D$. $\square$

Thus $T_L - T_R$ is the first intrinsic reflection-odd covariant of the interior profile. For $n = 3$ it vanishes identically. For the first two larger alphabets one obtains

$$\begin{aligned} n = 4, \quad g = (u, v): \qquad & T_L - T_R = uv(u - v), \\ n = 5, \quad g = (g_1, g_2, g_3): \qquad & T_L - T_R = g_2(g_1 - g_3)(g_1 + g_3). \end{aligned} \tag{10.8}$$

There is also a finite algebraic closure under repeated endpoint insertion. For a Laurent polynomial X, write $X^{\#}(z) := X(z^{-1})$. If

$$F = G + U, \qquad U = a + cz^N,$$

define the degree-1, degree-2, and degree-3 fields

$$G, \qquad A_G := GG^{\#}, \qquad B_G := G^2, \qquad C_G := G^2G^{\#}. \tag{10.9}$$

Proposition 10.3 (Closed Laurent jet)**.** *Endpoint insertion preserves the finite triangular family* (G, A_G, B_G, C_G)*. Explicitly,*

$$A_F = A_G + UG^{\#} + GU^{\#} + UU^{\#}, \tag{10.10}$$

$$B_F = B_G + 2UG + U^2, \tag{10.11}$$

$$C_F = C_G + 2UA_G + B_GU^{\#} + U^2G^{\#} + 2UGU^{\#} + U^2U^{\#}. \tag{10.12}$$

Thus repeated peeling creates new boundary evaluations but no new polynomial field of degree at most three.

Proof. The identities are the direct expansions of $FF^{\#}$, F^2, and $F^2F^{\#}$. $\square$

Differentiating (10.6) with respect to a and c gives an exact recursion for the endpoint cubic imbalance.

Proposition 10.4 (Recursive orientation equation)**.** *One has*

$$\mathcal{C}_f(0) = a^3 + 2aM_2 + 2ac^2 + cD + T_L, \tag{10.13}$$

$$\mathcal{C}_f(N) = c^3 + 2cM_2 + 2a^2c + aD + T_R, \tag{10.14}$$

and therefore

$$\begin{aligned} \mathcal{C}_f(0) - \mathcal{C}_f(N) = (a - c)\big(a^2 - ac + c^2 + 2M_2 - D\big) \\ + (T_L - T_R). \end{aligned} \tag{10.15}$$

The local coefficient is strictly positive unless $f = 0$, since $D \le M_2$.

Remark 10.5. At a critical point, (10.3) and (10.15) give

$$\lambda(a^{p-1} - c^{p-1}) = (a - c)\big(a^2 - ac + c^2 + 2M_2 - D\big) + (T_L - T_R).$$

Thus the sole inhomogeneous source of endpoint orientation is the same cubic boundary defect of the interior. This is the recursive object that must be estimated in a global reflection argument.

11. Support rigidity of exact extremizers

The positivity of the cubic field yields a strong classification of the support of an exact maximizer.

Proposition 11.1 (Support closure)**.** *Let $p > 1$ and let $f \geq 0$ be a global maximizer of Φ_p among profiles supported on I_n. If $f_k = 0$, then*

$$\mathcal{C}_f(k) = 0. \tag{11.1}$$

Consequently, with $A = \operatorname{supp} f$,

$$(A + A - A) \cap I_n \subseteq A. \tag{11.2}$$

Proof. If $f_k = 0$, perturb by $f + \varepsilon\delta_k$. The numerator has expansion

$$Q(f + \varepsilon\delta_k) = Q(f) + 4\varepsilon\mathcal{C}_f(k) + O(\varepsilon^2),$$

whereas the denominator changes by $O(\varepsilon^p)$. Since $p > 1$, maximality forces (11.1). All terms in $\mathcal{C}_f(k)$ are nonnegative, so no relation $a + b - c = k$ with $a, b, c \in A$ can occur outside A, proving (11.2). □

Theorem 11.2 (Arithmetic-progression support)**.** *Under the hypotheses of Proposition 11.1, the support of f is an arithmetic progression:*

$$\operatorname{supp} f = \{a, a + d, \dots, a + md\} \tag{11.3}$$

for some integers a, d, m with $d \geq 1$, unless f is a point mass.

Proof. Translate the support so that its minimum is 0, and let d be its least positive element. If x belongs to the support and $x + d$ does not exceed the maximum M, then closure gives $x + d = x + d - 0$ in the support. Hence all multiples of d up to M occur. If some support element y were not a multiple of d, write $y = qd + r$ with $0 < r < d$. Since qd is already in the support, closure gives $r = y + 0 - qd$ in the support, contradicting the minimality of d. □

Remark 11.3. Translation and integer dilation preserve all additive relations. Hence every nontrivial exact extremizer compresses to a positive full-support extremizer on a smaller alphabet. Arbitrary separated multi-packet supports cannot occur at exact equality; any boundary branch is a smaller alphabet branch in disguise.

12. Reflection geometry and the universal one-third law

Let $Jf(j) = f(N - j)$, and decompose

$$f = s + h, \qquad Js = s, \qquad Jh = -h. \tag{12.1}$$

The reflection invariance of Q forces an even polynomial in the odd component. In fact there is a stronger Fourier orthogonality.

Proposition 12.1 (Exact even–odd Fourier decomposition)**.** *For every real s, h satisfying (12.1) and every $\lambda \in \mathbb{R}$,*

$$\boxed{Q(s + \lambda h) = Q(s) + 2\lambda^2\|s * h\|_2^2 + \lambda^4 Q(h).} \tag{12.2}$$

Equivalently,

$$\langle s * s, h * h\rangle = -\|s * h\|_2^2. \tag{12.3}$$

Proof. After multiplying by the common phase $e^{\pi i N\xi}$, the Fourier transform of s is real-valued and that of h is purely imaginary. Consequently,

$$|\widehat{s + \lambda h}|^2 = |\widehat{s}|^2 + \lambda^2|\widehat{h}|^2.$$

Square and integrate. Comparison with the direct quartic expansion gives (12.3). □

This identity yields a global symmetrization principle in the Hilbert and super-Hilbert ranges.

Proposition 12.2 (Root-mean-square reflection symmetrization)**.** *Let $f \geq 0$ on I_n, and define the symmetric profile*

$$r_j := \left(\frac{f_j^2 + f_{N-j}^2}{2} \right)^{1/2}. \tag{12.4}$$

Then

$$Q(r) \geq Q(f). \tag{12.5}$$

For every $p \geq 2$,

$$\|r\|_p \leq \|f\|_p, \tag{12.6}$$

and hence

$$\Phi_p(f) \leq \Phi_p(r). \tag{12.7}$$

If f is positive on the full interval, equality forces $f = Jf$.

Proof. Write $s = (f + Jf)/2$, $h = (f - Jf)/2$, and $v_j = (s_j, h_j) \in \mathbb{R}^2$. For each lag k, the mixed symmetric–antisymmetric correlations cancel, so

$$R_f(k) = \sum_j \langle v_j, v_{j+k} \rangle.$$

Since $R_f(k) \geq 0$ and $r_j = |v_j|$, Cauchy–Schwarz gives

$$0 \leq R_f(k) \leq \sum_j |v_j||v_{j+k}| = R_r(k).$$

The autocorrelation formula $Q(f) = R_f(0)^2 + 2\sum_{k \geq 1} R_f(k)^2$ proves (12.5). The power-mean inequality on each reflection pair gives

$$f_j^p + f_{N-j}^p \geq 2 \left(\frac{f_j^2 + f_{N-j}^2}{2} \right)^{p/2},$$

which is (12.6). The equality statement follows from the strict cases of these inequalities. □

The range $4/3 < p < 2$ requires a different mechanism. We now prove the universal spectral theorem stated in the introduction.

Let $s_j > 0$ on I_n and assume $Js = s$. Define

$$R_s(k) := \sum_j s_j s_{j+k}, \qquad C_s(i) := \sum_j R_s(i-j) s_j = (s * s * \widetilde{s})(i). \tag{12.8}$$

The quantities in (1.17) satisfy

$$\sum_j P_s(i,j) = 1, \qquad \sum_i \pi_i = 1.$$

Lemma 12.3. *The kernel P_s is reversible with respect to π:*

$$\pi_i P_s(i,j) = \pi_j P_s(j,i). \tag{12.9}$$

Moreover,

$$\langle \phi, P_s \phi \rangle_{L^2(\pi)} = \frac{\|s * (s\phi)\|_2^2}{Q(s)} \geq 0. \tag{12.10}$$

Proof. Since R_s is even,

$$\pi_i P_s(i,j) = \frac{s_i s_j R_s(i-j)}{Q(s)} = \pi_j P_s(j,i).$$

Expanding the convolution norm gives

$$\|s * (s\phi)\|_2^2 = \sum_{i,j} R_s(i-j) s_i \phi_i s_j \phi_j,$$

which proves (12.10). □

Proof of Theorem 1.6. Put a probability distribution on additive quadruples by

$$\mathbb{P}(a,b,c,d) := \frac{s_a s_b s_c s_d}{Q(s)} \mathbf{1}_{\{a+b=c+d\}}. \tag{12.11}$$

Its (d,b) marginal is

$$\mathbb{P}(d,b) = \frac{s_d s_b R_s(d-b)}{Q(s)},$$

so the conditional law of b given d is $P_s(d,\cdot)$, and

$$\langle \phi, P_s \phi \rangle_{L^2(\pi)} = \mathbb{E}[\phi(d)\phi(b)]. \tag{12.12}$$

Let $m = N/2$ and set

$$Y_1 = a - m, \quad Y_2 = b - m, \quad Y_3 = m - c, \quad Y_4 = m - d.$$

Then $Y_1 + Y_2 + Y_3 + Y_4 = 0$. Reflection symmetry of s makes the joint law of the four Y_i exchangeable. If

$$\psi(y) := \phi(m+y),$$

then ψ is odd and

$$\phi(b) = \psi(Y_2), \qquad \phi(d) = -\psi(Y_4).$$

Write

$$A := \mathbb{E}\psi(Y_1)^2, \qquad B := \mathbb{E}[\psi(Y_1)\psi(Y_2)].$$

Exchangeability and (12.12) give

$$\|\phi\|_{L^2(\pi)}^2 = A, \qquad \langle \phi, P_s \phi \rangle_{L^2(\pi)} = -B.$$

Since a square is nonnegative,

$$0 \le \mathbb{E}\left(\sum_{r=1}^4 \psi(Y_r)\right)^2 = 4A + 12B.$$

Thus $-B \le A/3$. The lower bound follows from Lemma 12.3.

For $\psi(y) = y$, the sum in the preceding display vanishes identically, so equality holds. If s has full positive support and equality holds for a general odd ψ, then

$$\phi(a) + \phi(b) = \phi(c) + \phi(d) \qquad \text{whenever } a + b = c + d.$$

Hence $\phi(a) + \phi(b)$ depends only on $a + b$. Taking neighboring sums shows that $\phi_{j+1} - \phi_j$ is independent of j, so ϕ is affine. Oddness about $N/2$ leaves only scalar multiples of $j - N/2$. □

12.1. **The reflection Hessian.** Suppose now that s is a positive symmetric critical point of Φ_p. Normalize

$$S_p := \sum_i s_i^p.$$

The Euler–Lagrange equation gives $C_s(i) = \lambda s_i^{p-1}$, and Euler's identity gives $Q(s) = \lambda S_p$. Consequently,

$$\pi_i = \frac{s_i^p}{S_p}. \tag{12.13}$$

Let $h_i = s_i\phi_i$ with ϕ reflection odd. The first variation vanishes. Expanding the numerator by Proposition 12.1 and the denominator by the binomial theorem yields the exact second-variation formula

$$\frac{\Phi_p''(s)[h,h]}{4\Phi_p(s)} = \langle \phi, P_s\phi\rangle_{L^2(\pi)} - (p-1)\|\phi\|_{L^2(\pi)}^2. \tag{12.14}$$

Combining this with Theorem 1.6 gives the following.

Corollary 12.4 (Universal reflection stability)**.** *At every positive symmetric critical point,*

$$\frac{\Phi_p''(s)[s\phi,s\phi]}{4\Phi_p(s)} \le -\left(p-\frac{4}{3}\right)\|\phi\|_{L^2(\pi)}^2 \tag{12.15}$$

for every reflection-odd ϕ. Thus:

(i) *if $p > 4/3$, all nonzero odd directions are strictly decreasing;*
(ii) *if $p = 4/3$, the only second-order neutral odd direction is $h_j = (j-N/2)s_j$;*
(iii) *if $1 < p < 4/3$, this linear-tilt direction is increasing.*

The local spectral law can be made finite-radius around a symmetric global maximizer.

Proposition 12.5 (A quantitative reflection basin)**.** *Let $4/3 < p < 2$, and let $s > 0$ be a symmetric global maximizer of Φ_p, normalized by $\sum_i s_i^p = 1$. Let*

$$h_i = s_i\phi_i, \qquad J\phi = -\phi, \qquad |\phi_i| \le 1,$$

and set

$$W := \sum_i s_i^p\phi_i^2. \tag{12.16}$$

If

$$W \le 2\left(p-\frac{4}{3}\right), \tag{12.17}$$

then

$$\Phi_p(s+h) \le \Phi_p(s). \tag{12.18}$$

Proof. By Theorem 1.6 and Proposition 12.1,

$$\frac{Q(s+h)}{Q(s)} \le 1 + \frac{2}{3}W + \frac{Q(h)}{Q(s)}.$$

Global maximality of s gives $Q(h) \le Q(s)\|h\|_p^4$. Since $p < 2$ and the weights s_i^p form a probability measure,

$$\|h\|_p^4 = \left(\sum_i s_i^p|\phi_i|^p\right)^{4/p} \le W^2.$$

Therefore

$$\frac{Q(s+h)}{Q(s)} \le 1 + \frac{2}{3}W + W^2. \tag{12.19}$$

For $|x| \le 1$ and $1 < p < 2$,

$$\frac{(1+x)^p + (1-x)^p}{2} \ge 1 + \frac{p(p-1)}{2} x^2.$$

Pairing reflected coordinates and using Bernoulli's inequality gives

$$\|s+h\|_p^4 \ge 1 + 2(p-1)W. \tag{12.20}$$

The condition (12.17) makes the right side of (12.19) no larger than that of (12.20). ☐

Remark 12.6. In the conjectural large-alphabet critical window $p - 4/3 \asymp 1/\log n$, the basin has weighted squared radius of order $1/\log n$. Thus a broad inverse theorem need only bring a near-extremizer into this neighborhood; local reflection stability would then finish the symmetry step.

13. Exact parity renormalization

Let

$$e_j := f_{2j}, \qquad o_j := f_{2j+1}. \tag{13.1}$$

Then

$$(f * f)_{2m} = (e * e)_m + (o * o)_{m-1}, \qquad (f * f)_{2m+1} = 2(e * o)_m. \tag{13.2}$$

Let $\tau v(m) = v(m-1)$ and put

$$M := \|e * o\|_2^2, \qquad H := \langle e * e, \tau(o * o)\rangle. \tag{13.3}$$

The coefficients are nonnegative, and Fourier Cauchy–Schwarz gives

$$0 \le H \le M \le \sqrt{Q(e)Q(o)}. \tag{13.4}$$

Set

$$x := Q(e)^{1/4}, \qquad y := Q(o)^{1/4}, \qquad \rho := \frac{M}{x^2y^2}, \qquad \kappa := \frac{H}{M} \tag{13.5}$$

when $xyM > 0$, with the evident continuous conventions otherwise. Define

$$\Delta_{\text{ren}}(e, o) := 6(1-\rho) + 2\rho(1-\kappa). \tag{13.6}$$

Proof of Theorem 1.7. By (13.2),

$$Q(f) = Q(e) + Q(o) + 4M + 2H. \tag{13.7}$$

Substituting (13.5) gives (1.20).

For the square representation, let $E = \widehat{e}$, $O = \widehat{o}$, and $O_{1/2} = e^{-\pi i \xi} O$. First,

$$3 \left\| \frac{|E|^2}{x^2} - \frac{|O|^2}{y^2} \right\|_2^2 = 6(1-\rho).$$

Next, with $z = E\overline{O_{1/2}}$,

$$\|z - \overline{z}\|_2^2 = 2M - 2 \operatorname{Re} \int_{\mathbb{T}} e^{2\pi i \xi} E(\xi)^2 \overline{O(\xi)^2} \, d\xi = 2(M - H).$$

Dividing by x^2y^2 gives $2\rho(1-\kappa)$ and proves (1.21). ☐

The square formula gives a concrete interpretation of small defect: $|E|^2/x^2$ and $|O|^2/y^2$ must be close, and the two polyphase components must have nearly compatible half-lattice phase.

13.1. **The local gain function.** Let

$$t := \frac{4}{p}, \qquad u := \frac{x^p}{x^p + y^p}.$$

From (1.20),

$$\frac{Q(f)}{(x^p + y^p)^{4/p}} = B_t(u, \Delta), \tag{13.8}$$

where

$$B_t(u, \Delta) := u^t + (1-u)^t + (6 - \Delta)[u(1-u)]^{t/2}. \tag{13.9}$$

When $t \geq 2$ (equivalently, $p \leq 2$), combining this with the sharp two-point envelope gives

$$B_t(u, \Delta) \leq 1 + \big(8 - 2^t - \Delta\big)[u(1-u)]^{t/2}. \tag{13.10}$$

Thus a sufficient one-step closure condition is

$$\Delta_{\mathrm{ren}}(e, o) \geq 8 - 2^{4/p}. \tag{13.11}$$

Near $p = 4/3$,

$$8 - 2^{4/p} = 18(\log 2)\left(p - \frac{4}{3}\right) + O\left(\left(p - \frac{4}{3}\right)^2\right). \tag{13.12}$$

This identifies the required defect budget per active scale in the critical window.

13.2. **The exact multiplicative cascade.** Define

$$\mathcal{Z}_p(f) := Q(f)^{p/4} = Q(f)^{1/t}. \tag{13.13}$$

Since $x^p = \mathcal{Z}_p(e)$ and $y^p = \mathcal{Z}_p(o)$, (13.8) is equivalent to

$$\mathcal{Z}_p(f) = \big(\mathcal{Z}_p(e) + \mathcal{Z}_p(o)\big) B_t(u, \Delta)^{1/t}. \tag{13.14}$$

Build the dyadic parity tree by repeatedly taking even and odd children. At a node v, let

$$\beta_v := B_t(u_v, \Delta_v)^{1/t}. \tag{13.15}$$

After sufficiently many generations, the leaves are individual coefficients of f. Expanding (13.14) gives the following exact identity.

Theorem 13.1 (Multiplicative cascade)**.** *For every finitely supported f and every $p > 0$,*

$$\boxed{\mathcal{Z}_p(f) = \sum_j |f_j|^p \prod_{v \prec j} \beta_v,} \tag{13.16}$$

where the product is over the internal nodes on the path from the root to the leaf containing j. Equivalently,

$$\Phi_p(f)^{p/4} = \sum_j \frac{|f_j|^p}{\|f\|_p^p} \prod_{v \prec j} \beta_v. \tag{13.17}$$

Proof. Equation (13.14) writes the value at each node as a local gain times the sum of its two children. Iterating this identity to the singleton leaves gives (13.16); division by $\|f\|_p^p$ gives (13.17). ☐

Thus the global sharp inequality becomes a pathwise multiscale question: with a leaf chosen according to normalized p-mass, the expected product of the local gains must not exceed 1.

13.3. **A one-scale atomic–smooth dichotomy.** At the Euclidean exponent $t = 3$ and zero defect,

$$B_3(u,0) = 1 - G(u), \tag{13.18}$$

where

$$G(u) := 3u(1-u)\big(1 - 2\sqrt{u(1-u)}\big) \geq 0. \tag{13.19}$$

The zero set is exactly $\{0, 1/2, 1\}$. Moreover,

$$G(u) \asymp \begin{cases} u, & u \to 0, \\ 1-u, & u \to 1, \\ (u-1/2)^2, & u \to 1/2. \end{cases} \tag{13.20}$$

Uniform boundedness of the t-derivatives on a compact neighborhood of $t = 3$ gives the following quantitative statement.

Proposition 13.2 (One-scale dichotomy)**.** *There are absolute constants $C, c > 0$ such that, whenever $|t-3| \leq c$, $0 \leq \Delta \leq 6$, and*

$$B_t(u,\Delta) \geq 1 - \eta, \tag{13.21}$$

one has

$$G(u) + \Delta[u(1-u)]^{t/2} \leq C(|3-t| + \eta). \tag{13.22}$$

Consequently, one of the following holds:

(i) atomic split: $u \leq C(|3-t|+\eta)$ *or* $1 - u \leq C(|3-t|+\eta)$*;*

(ii) balanced smooth split:

$$\left|u - \frac{1}{2}\right| \leq C\sqrt{|3-t|+\eta}, \qquad \Delta \leq C(|3-t|+\eta).$$

Proof. The functions u^t, $(1-u)^t$, and $[u(1-u)]^{t/2}$ have uniformly bounded t-derivatives for t in a fixed compact subinterval of $(2,\infty)$, including at $u = 0, 1$ by continuity. Hence

$$B_t(u,\Delta) \leq 1 - G(u) + C|3-t| - \Delta[u(1-u)]^{t/2}.$$

This proves (13.22). The alternatives then follow from (13.20); in the balanced regime, $u(1-u)$ is bounded below by an absolute positive constant, so the defect term also controls Δ. □

In the predicted window $3 - t \asymp 1/\log n$, every non-atomic near-lossless node must therefore satisfy

$$\left|u - \frac{1}{2}\right| \lesssim \frac{1}{\sqrt{\log n}}, \qquad \Delta_{\mathrm{ren}} \lesssim \frac{1}{\log n}.$$

This is a rigorous one-scale form of the atomic/Gaussian dichotomy. What remains is to propagate it along most active paths and most active scales.

14. The discrete–continuous bridge

There is a second, complementary route to the large-alphabet problem. For a finite sequence b, define its hat-function interpolation by

$$Ib(x) := \sum_{k \in \mathbb{Z}} b_k (1 - |x-k|)_+. \tag{14.1}$$

Thus $Ib(k) = b_k$, the function is affine on each unit interval, and it vanishes one unit beyond the extreme nonzero knots.

Lemma 14.1. *For every finitely supported sequence b,*

$$\boxed{\|b\|_{\ell^2}^2 = \|Ib\|_{L^2(\mathbb{R})}^2 + \frac{1}{6}\|\Delta b\|_{\ell^2}^2,} \tag{14.2}$$

where $\Delta b(k) = b(k+1) - b(k)$.

Proof. On each unit interval, integrate the square of the affine function joining b_k and b_{k+1}. Summing gives

$$\|Ib\|_2^2 = \frac{2}{3}\sum_k b_k^2 + \frac{1}{3}\sum_k b_k b_{k+1}.$$

The difference from $\sum_k b_k^2$ is $\frac{1}{6}\sum_k (b_{k+1} - b_k)^2$. □

Iterating (14.2) gives a positive square hierarchy.

Corollary 14.2 (Difference hierarchy)**.** *For every integer $M \geq 1$,*

$$\|b\|_2^2 = \sum_{j=0}^{M-1} 6^{-j}\|I\Delta^j b\|_2^2 + 6^{-M}\|\Delta^M b\|_2^2. \tag{14.3}$$

The remainder tends to zero as $M \to \infty$. For $b = f * f$,

$$\Delta^{2m}(f * f) = (\Delta^m f) * (\Delta^m f), \tag{14.4}$$

$$\Delta^{2m+1}(f * f) = (\Delta^{m+1} f) * (\Delta^m f). \tag{14.5}$$

Proof. Iterate Lemma 14.1. Since $\|\Delta\|_{\ell^2 \to \ell^2} \leq 2$, the remainder is at most $(4/6)^M\|b\|_2^2$. The convolution identities follow from the Fourier multiplier of Δ. □

14.1. **A scale-dependent embedding.** Let $f = (f_0, \ldots, f_{n-1})$, and for $p > 0$ define the step function

$$G_{n,p}(x) := n^{1/p} f_j, \qquad x \in [j/n, (j+1)/n). \tag{14.6}$$

Then

$$\|G_{n,p}\|_{L^p(\mathbb{R})} = \|f\|_{\ell^p}. \tag{14.7}$$

If $b = f*f$, the convolution $G_{n,p}*G_{n,p}$ is, up to scaling and a translation, the piecewise-linear interpolation of b. Therefore Lemma 14.1 gives an exact discrete–continuous bridge.

Proposition 14.3 (Exact scale-n bridge)**.** *For every finite profile f supported in I_n,*

$$\boxed{Q(f) = n^{3-4/p}\|G_{n,p} * G_{n,p}\|_{L^2(\mathbb{R})}^2 + \frac{1}{6}\|\Delta(f * f)\|_{\ell^2}^2.} \tag{14.8}$$

Proof. Let $T = \mathbf{1}_{[0,1]} * \mathbf{1}_{[0,1]}$. If $b = f * f$, then

$$(G_{n,p} * G_{n,p})(x) = n^{2/p-1}\sum_s b_s T(nx - s).$$

The sum on the right is a translate of the hat-function interpolation Ib. Hence

$$\|G_{n,p} * G_{n,p}\|_2^2 = n^{4/p-3}\|Ib\|_2^2.$$

Substitute this identity into Lemma 14.1. □

Let

$$\Gamma := \frac{3\sqrt{3}}{4}, \qquad Y^2 := \Gamma^{-1} = \frac{4}{3\sqrt{3}}. \tag{14.9}$$

The sharp continuous Young inequality at exponent $4/3$ is

$$\|g * g\|_2^2 \leq Y^2\|g\|_{4/3}^4, \tag{14.10}$$

with Gaussian extremizers [1, 6]. Define the continuous deficit

$$\mathrm{Def}_Y(g) := Y^2\|g\|_{4/3}^4 - \|g * g\|_2^2 \geq 0. \tag{14.11}$$

Since

$$\|f\|_{4/3}^4 = n^{3-4/p}\|G_{n,p}\|_{4/3}^4,$$

(14.8) becomes

$$\boxed{Q(f) = Y^2\|f\|_{4/3}^4 - n^{3-4/p}\,\mathrm{Def}_Y(G_{n,p}) + \frac{1}{6}\|\Delta(f*f)\|_2^2.} \tag{14.12}$$

Thus the desired inequality $Q(f) \le \|f\|_p^4$ is exactly equivalent to

$$\frac{1}{6}\|\Delta(f*f)\|_2^2 - n^{3-4/p}\,\mathrm{Def}_Y(G_{n,p}) \le \|f\|_p^4 - Y^2\|f\|_{4/3}^4. \tag{14.13}$$

The positive lattice roughness and the positive Euclidean Young deficit appear with opposite signs. Atomic profiles have large contributions of both types; broad Gaussian profiles have both contributions small.

15. Entropy formulation and the two-well picture

For $p > 4/3$, put

$$\rho := \frac{3p}{4} > 1, \qquad \delta := 3 - \frac{4}{p} = \frac{3(\rho-1)}{\rho}. \tag{15.1}$$

For $f \ne 0$, define a probability distribution

$$\mu_j := \frac{f_j^{4/3}}{\sum_k f_k^{4/3}} \tag{15.2}$$

and its Rényi entropy

$$H_\rho(\mu) := \frac{1}{1-\rho}\log\sum_j \mu_j^\rho. \tag{15.3}$$

A direct calculation gives the exact identity

$$\boxed{\frac{\|f\|_p^4}{\|f\|_{4/3}^4} = \exp\big(-\delta H_\rho(\mu_f)\big).} \tag{15.4}$$

Define the normalized additive energy

$$\mathscr{E}(f) := \frac{Q(f)}{\|f\|_{4/3}^4}. \tag{15.5}$$

Then $Q(f) \le \|f\|_p^4$ is equivalent to

$$-\log\mathscr{E}(f) \ge \delta H_\rho(\mu_f). \tag{15.6}$$

Accordingly, set

$$\Lambda_n(\rho) := \inf_{\substack{0\ne f:\, I_n\to[0,\infty)\\ H_\rho(\mu_f)>0}} \frac{-\log\mathscr{E}(f)}{H_\rho(\mu_f)}. \tag{15.7}$$

The exponent p is admissible exactly when

$$\delta \le \Lambda_n(\rho). \tag{15.8}$$

This formulation makes the large-alphabet constant transparent. A broad Gaussian profile has

$$-\log\mathscr{E}(f) \sim \log\Gamma, \qquad H_\rho(\mu_f) \sim \log n,$$

whereas a point mass has both entropy and energy loss equal to zero. Thus the predicted first-order asymptotic is equivalent to

$$\Lambda_n(\rho) = (1+o(1))\frac{\log\Gamma}{\log n} \tag{15.9}$$

in the window $\rho = 1 + O(1/\log n)$.

The correct inverse statement therefore has two wells rather than one:

$$\boxed{\begin{array}{c}\text{near extremizer} \Longrightarrow \\ \text{atomic profile} \quad \text{or} \quad \text{broad approximately Gaussian profile}\end{array}} \tag{15.10}$$

The support theorem rules out arbitrary exact multi-packets, while the one-scale renormalization theorem rules out a near-lossless intermediate split unless it is either highly atomic or balanced with small phase and magnitude defect.

16. The remaining multiscale rigidity problem

Let t_n denote the optimal dimension-free additive-energy exponent on $\{0, 1, \dots, n-1\}^d$. Shao proved nontrivial upper and lower bounds and the Gaussian lower side predicts

$$t_n = 3 - (1 + o(1)) \log_n \Gamma, \qquad \Gamma = \frac{3\sqrt{3}}{4}. \tag{16.1}$$

Equivalently,

$$\lim_{n\to\infty} (3 - t_n) \log_2 n = \log_2 \Gamma = 0.3774437510\dots. \tag{16.2}$$

This paper does not prove (16.1). The preceding exact results reduce the missing argument to the following multiscale rigidity problem.

Conjecture 16.1 (Refinement rigidity)**.** *Let $p_n = 4/3 + O(1/\log n)$ and let $f^{(n)}$ be normalized profiles whose sharp quotients are asymptotically critical. Outside paths carrying asymptotically atomic mass, the parity cascade satisfies, on all but $o(\log n)$ active scales,*

$$\left| u_v - \frac{1}{2} \right| = o(1), \qquad \Delta_v = o(1).$$

After translation and rescaling, the non-atomic component is precompact in a continuum topology strong enough to pass to Young's inequality. Every nonzero limit is a Gaussian extremizer.

A proof should naturally divide into three stages.

1. *Cascade concentration.* Use (13.17) and Proposition 13.2 to show that a near-critical average path has only atomic or balanced-small-defect nodes on most active scales.
2. *Refinement compactness.* Convert the square defect (1.21) into quantitative compatibility of neighboring dyadic polyphase components and hence into regularity of a rescaled interpolation.
3. *Gaussian identification.* Combine the compactness with the compensated identity (14.12), the Euler–Lagrange equation, and stability of the sharp continuous Young inequality to identify the limit.

The reflection theorem is now local rather than global: once the broad profile enters the $O(1/\log n)$ basin of a symmetric branch, Proposition 12.5 closes the odd directions automatically. The remaining task is therefore not an n-variable symmetrization, but a multiscale compactness and rigidity theorem.

17. Beyond the Hilbertian point

The exact tensorisation in Proposition 3.1 is valid for every $r \geq 1$, even though the one-dimensional optimisation is explicitly solved here only for $r = 2$. It is useful to isolate the obstruction obtained by taking one input constant. For $1 \leq r < \infty$ and $q \in [1, \infty]$, define

$$\Gamma_r(q) := \sup_{0\leq t\leq 1} \left[\frac{1}{r} \log_2\big(1 + (1+t)^r + t^r\big) - \frac{1}{q} \log_2(1 + t^q) \right], \tag{17.1}$$

where the second term is interpreted as zero for $q = \infty$.

Proposition 17.1. *Let $p, q \in [1, \infty]$ and $1 \le r < \infty$. If*

$$\|f * g\|_r \le \|f\|_p \|g\|_q \tag{17.2}$$

holds for every dimension and all functions supported on Q_d, then

$$\frac{1}{p} \ge \Gamma_r(q), \qquad \frac{1}{q} \ge \Gamma_r(p). \tag{17.3}$$

For $r = 2$, these conditions are also sufficient.

Proof. By exact tensorisation, (17.2) is equivalent to the two-point constant being at most one. Taking the first input to be $(1, 1)$ and the second to be $(1, t)$ gives

$$\big(1 + (1 + t)^r + t^r\big)^{1/r} \le 2^{1/p}(1 + t^q)^{1/q},$$

which is equivalent to the first condition in (17.3). Interchanging the inputs gives the second. When $r = 2$,

$$\Gamma_2(q) = \frac{1 + \Lambda(q)}{2},$$

so sufficiency is precisely Corollary 1.2. □

The choices $t = 0$ and $t = 1$ in (17.1) recover the familiar necessary conditions

$$p, q \le r, \qquad \frac{1}{p} + \frac{1}{q} \ge \frac{\log_2(2 + 2^r)}{r}.$$

The diagonal problem $p = q$ is solved for every $r \ge 1$ in [4, 7]. The endpoint $r = 1$ is elementary: the conditions above force $p = q = 1$. For $1 < r < \infty$ with $r \ne 2$, a complete off-diagonal characterisation is not known beyond the diagonal and partial off-diagonal results in [4, 7]. The special feature of $r = 2$ is the factorisation (4.1): after ℓ^2 normalisation, the numerator depends on the two inputs only through the product uv. For a general r, the expression

$$1 + (x + y)^r + (xy)^r$$

retains genuinely two-variable dependence, and interior extremisers are not excluded by the present argument. Nevertheless, Propositions 3.1 and 17.1 reduce the problem completely to an explicit two-variable extremal problem and identify the exact boundary tests that become sufficient in the quadratic case.

For later compactness arguments it is useful to separate the point-mass tails from the compact interior. In anti-aligned point-mass coordinates set

$$\mathcal{R}_{p,q,r}(x, y) := \frac{\big[x^r + y^r + (1 + xy)^r\big]^{1/r}}{(1 + x^p)^{1/p}(1 + y^q)^{1/q}}, \qquad 0 \le x, y \le 1. \tag{17.4}$$

Lemma 17.2. *Let K be a compact set of finite exponent triples (p, q, r) satisfying*

$$1 \le p < r, \quad 1 \le q < r, \quad \frac{1}{p} + \frac{1}{q} > 1 \tag{17.5}$$

at every point of K. There is $\rho \in (0, 1)$ such that, uniformly for $(p, q, r) \in K$,

$$\mathcal{R}_{p,q,r}(x, y) < 1$$

whenever $(x, y) \ne (0, 0)$ and $\min\{x, y\} < \rho$.

The three strict conditions are sharp for this type of tail exclusion. If $p \ge r$, then $\mathcal{R}_{p,q,r}(x, 0) \ge 1$ for $x > 0$, with equality for $p = r$ and strict inequality for $p > r$; the analogous assertion holds for q and y. If $a := 1/p + 1/q \le 1$, then along $x = s^{1/p}$, $y = s^{1/q}$ one has $\mathcal{R}_{p,q,r}(x, y) > 1$ for every $s \in (0, 1]$ when $a = 1$, and for all sufficiently small $s > 0$ when $a < 1$.

Proof. For $0 < t \le 1$ put $h_t(c) = c^{-1}\log(1+t^c)$. Direct differentiation shows that $h_t'(c) < 0$. Hence, for example,

$$\mathcal{R}_{p,q,r}(0,y) = \frac{(1+y^r)^{1/r}}{(1+y^q)^{1/q}} < 1$$

when $0 < y \le 1$ and $q < r$, and similarly on the other punctured axis.

We prove a uniform corner estimate. Compactness and strictness give

$$\begin{aligned} M &:= \max_K \max\{p,q,r\} < \infty, \\ d &:= \min_K \min\{r-p, r-q\} > 0, \\ \sigma &:= \min_K \left(\frac{1}{p} + \frac{1}{q} - 1\right) > 0, \end{aligned}$$

and let $m := \min_K \min\{p,q\} \ge 1$ and $C := M2^{M-1}$. If $0 \le x, y \le \delta \le 1$, write $X = x^p$, $Y = y^q$, and $T = X + Y$. The estimates $\log(1+v) \le v$, $\log(1+v) \ge v/2$ for $0 \le v \le 1$, and $(1+xy)^r - 1 \le Cxy$ give

$$\log \mathcal{R}_{p,q,r}(x,y) \le x^r + y^r + Cxy - \frac{T}{2M}.$$

Here $x^r \le \delta^d X$ and $y^r \le \delta^d Y$. Moreover, if $b = 1/p + 1/q$, then

$$xy = X^{1/p}Y^{1/q} \le T^b \le T(2\delta^m)^\sigma$$

once $2\delta^m \le 1$. Thus

$$\log \mathcal{R}_{p,q,r}(x,y) \le \left[\delta^d + C(2\delta^m)^\sigma - \frac{1}{2M}\right] T.$$

Choosing δ so that the bracket is negative proves strict inequality on $[0,\delta]^2 \setminus \{(0,0)\}$. On the compact sets $K \times \{0\} \times [\delta,1]$ and $K \times [\delta,1] \times \{0\}$, the punctured-axis inequalities have a uniform strict gap. Uniform continuity extends them to strips of widths $\eta_x, \eta_y > 0$. Taking $\rho = \min\{\delta, \eta_x, \eta_y\}$ covers every point in the asserted tail.

For sharpness, the axis formula and the strict decrease of h_x give the claims when $p \ge r$ or $q \ge r$. Finally, on the displayed path with $a \le 1$ the denominator is $(1+s)^a$, while the numerator is strictly larger than $1 + s^a$. This proves the claim for $a = 1$; for $a < 1$, it follows for small s from

$$\frac{\log(1+s^a) - a\log(1+s)}{s^a} \longrightarrow 1.$$

□

Remark 17.3. The lemma removes a noncompact tail for compact exponent families satisfying (17.5). It does not compare critical points in the remaining compact region and therefore does not by itself prove global fixed-q sharpness for $r \ne 2$.

17.1. **Signed hyperbolic coordinates and orientation.** The two relative orientations of two strictly positive non-flat vectors are degenerate when $r = 2$, and are strictly separated when $r > 2$. Equivalently, this assertion concerns two finite nonzero hyperbolic biases. To make it precise, put

$$f_z = (e^{z/2}, e^{-z/2}), \qquad g_w = (e^{-w/2}, e^{w/2}). \tag{17.6}$$

Homogeneity and a common reflection of the two coordinates show that positive two-vectors are covered by $z \in \mathbb{R}$ and $w \ge 0$, with zero coordinates obtained as limits. Their convolution is

$$f_z * g_w = \left(e^{(z-w)/2}, 2\cosh\frac{z+w}{2}, e^{(-z+w)/2}\right).$$

For finite p, q, r, define the logarithmic quotient

$$\begin{aligned}\Phi_{p,q,r}(z,w) := \frac{1}{r}\log\left[2\cosh\frac{r(z-w)}{2} + \left(2\cosh\frac{z+w}{2}\right)^r\right] \\ - \frac{1}{p}\log\left(2\cosh\frac{pz}{2}\right) - \frac{1}{q}\log\left(2\cosh\frac{qw}{2}\right). \qquad (17.7)\end{aligned}$$

Lemma 17.4. *Let $r \geq 2$. Among the two relative orientations of two nonnegative two-vectors with prescribed unordered coordinate magnitudes, the ℓ^r norm of the convolution is maximised when the larger coordinates are oppositely placed. For $r = 2$ the two orientations always give the same norm. For $r > 2$, equality holds if and only if at least one vector is flat or a point mass; equivalently, the inequality is strict if and only if both vectors have two strictly positive, unequal coordinates.*

Proof. Write the two biases as $z, w \geq 0$ and set

$$J_r(a) := \left(2\cosh\frac{a}{2}\right)^r - 2\cosh\frac{ra}{2}.$$

The anti-aligned numerator to the power r minus the aligned numerator to the power r is

$$J_r(z+w) - J_r(|z-w|).$$

Moreover,

$$J_r'(a) = r\left[\left(2\cosh\frac{a}{2}\right)^{r-1}\sinh\frac{a}{2} - \sinh\frac{ra}{2}\right]. \qquad (17.8)$$

It remains to prove

$$\frac{\sinh(rs)}{\sinh s} \leq (2\cosh s)^{r-1}, \qquad s > 0, \quad r \geq 2. \qquad (17.9)$$

With $x = e^{-2s}$, this is equivalent to

$$1 - x^r \leq (1-x)(1+x)^{r-1}. \qquad (17.10)$$

The logarithm of the right-hand side divided by the left-hand side vanishes at $r = 2$, and its r-derivative is

$$\log(1+x) + \frac{x^r \log x}{1-x^r} \geq \log(1+x) + \frac{x^2\log x}{1-x^2}.$$

The last expression is nonnegative because

$$K(x) := (1-x^2)\log(1+x) + x^2\log x$$

vanishes at 0 and 1 and is strictly concave on $(0,1)$:

$$K''(x) = 2\left[\log\frac{x}{1+x} + \frac{1}{1+x}\right] < 0.$$

Thus (17.10) holds, strictly for $r > 2$. It follows from (17.8) that J_r is nondecreasing, and it is strictly increasing on $(0,\infty)$ when $r > 2$. Since $z + w > |z - w|$ exactly when $z, w > 0$, the stated strictness follows for finite positive biases. Flat vectors and point masses give equality: if the ordered magnitudes are $A \geq B \geq 0$ and $C \geq D \geq 0$, then $A = B$ or $B = 0$ makes the two convolution triples permutations of one another, and the same is true if $C = D$ or $D = 0$. These are all the remaining cases, proving the equality statement. Finally $J_2 \equiv 2$. ∎

At the quadratic curved boundary one of the inputs is flat, so the two orientations coincide. The next calculation shows that this edge profile is transversely unstable as soon as $r > 2$.

Proposition 17.5 (Transverse instability)**.** *For every finite p, q, every $w > 0$, and every $r > 2$,*

$$\partial_z \Phi_{p,q,r}(0, w) = \frac{(2\cosh(w/2))^{r-1}\sinh(w/2) - \sinh(rw/2)}{2\cosh(rw/2) + (2\cosh(w/2))^r} > 0. \tag{17.11}$$

At $r = 2$ the derivative vanishes identically.

Consequently, for each fixed $q \in (1, 4/3)$ there is $\varepsilon_q > 0$ such that, whenever $2 < r < 2+\varepsilon_q$ and

$$\frac{1}{p_e(r,q)} = \Gamma_r(q),$$

both conditions in (17.3) hold at $(p_e(r,q), q)$, but

$$\mathsf{C}_1(p_e(r,q), q; r) > 1.$$

Thus the two edge conditions cease to be jointly sufficient immediately to the right of $r = 2$.

Proof. Differentiating (17.7) at $z = 0$ gives (17.11); its sign is the strict form of (17.9). Fix now $q \in (1, 4/3)$. At $r = 2$, the maximiser defining $\Gamma_2(q)$ is the unique interior point t_q from Lemma 5.1. Its second derivative is negative. Choose a small neighbourhood U of t_q, shrinking it if necessary so that it lies inside the IFT uniqueness neighbourhood. The continuous edge objective at $r = 2$ has a strict positive gap between its value at t_q and its maximum on $[0,1] \setminus U$. Since the edge objectives converge uniformly on $[0,1]$ as $r \to 2$, every global maximiser for nearby r lies in U. The implicit analytic implicit function theorem and the nonzero second derivative give a unique real-analytic critical point there. Every global maximiser lies in U and is therefore that point. Thus the global edge maximiser, its positive bias $w_e(r,q)$, and $\Gamma_r(q)$ are real analytic for r near 2. At the edge exponent,

$$\Phi_{p_e,q,r}(0, w_e) = 0,$$

while (17.11) supplies a direction with positive derivative. Hence the two-variable supremum is positive.

It remains to check the other edge condition. Write $p_0 = 1/\Gamma_2(q)$. The elementary estimates

$$\sqrt{2(1+t+t^2)} \le \sqrt{2}(1+t), \qquad (1+t^q)^{1/q} \ge 2^{1/q-1}(1+t)$$

give $\Gamma_2(q) \le 3/2 - 1/q < 3/4$, hence $p_0 > 4/3$. Since the interior maximiser t_q strictly beats $t = 1$,

$$\Lambda(q) + \frac{2}{q} > \log_2 3.$$

Using the explicit formula in Lemma 5.1 at $p_0 > 4/3$ and $2/p_0 = 1 + \Lambda(q)$, we obtain

$$\frac{1}{q} - \Gamma_2(p_0) = \frac{\Lambda(q) + 2/q - \log_2 3}{2} > 0.$$

The second edge condition therefore persists by continuity after decreasing ε_q. ◻

17.2. **The fixed-q global branch.** We next quantify the bifurcation. It is convenient to write $\alpha = 1/p$ and to regard (17.7) as $\Phi_{\alpha,q,r}$, so that the first norm term is

$$-\alpha \log\left(2\cosh\frac{z}{2\alpha}\right).$$

Lemma 17.6. *Put*

$$N_r(z,w) := 2\cosh\frac{r(z-w)}{2} + \left(2\cosh\frac{z+w}{2}\right)^r.$$

When $r = 2$,

$$N_2(z,w) = 2\big(1 + 2\cosh z\cosh w\big). \tag{17.12}$$

At every $(0,w)$ *one has*

$$\Phi_z = 0, \quad \Phi_{zz} = \frac{\cosh w}{1+2\cosh w} - \frac{p}{4}, \quad \Phi_{zw} = \Phi_{z\alpha} = \Phi_{w\alpha} = 0, \quad \Phi_\alpha = -\log 2. \tag{17.13}$$

If $t = e^{-w}$, $P(t) = 1 + t + t^2$, *and*

$$K_q(t) := t^{q-1}(2+t) - (1+2t),$$

then

$$\Phi_w(0,w) = \frac{tK_q(t)}{2P(t)(1+t^q)}. \tag{17.14}$$

At a zero of K_q,

$$\Phi_{ww}(0,w) = -\frac{t^2 K_q'(t)}{2P(t)(1+t^q)}. \tag{17.15}$$

Finally,

$$\Phi_{zr}(0,w)|_{r=2} = \frac{\sinh w \log(2\cosh(w/2)) - (w/2)\cosh w}{2+4\cosh w}. \tag{17.16}$$

Proof. The addition formulas for cosh give (17.12). Differentiating its logarithm and the two norm terms gives (17.13). At $z = 0$, cancellation of the $\log t$ terms yields

$$\Phi(0,w) = \left(\frac{1}{2} - \alpha\right)\log 2 + \frac{1}{2}\log P(t) - \frac{1}{q}\log(1+t^q).$$

Since $\mathrm{d}t/\mathrm{d}w = -t$, this proves (17.14); differentiating once more at a zero of K_q proves (17.15). The numerator of $\Phi_z(0,w)$ vanishes at $r = 2$. Differentiating that numerator in r and dividing by its $r = 2$ denominator gives (17.16). □

The following exact quadratic equality set will be a useful base point for any future local-to-global continuation.

Proposition 17.7 (Quadratic equality set and compact gap)**.** *Fix* $q \in (1, 4/3)$, *let* $t_0 = t_q$, *and put* $p_0 = 1/\Gamma_2(q)$. *Then* $4/3 < p_0 < 2$ *and*

$$\mathcal{R}_{p_0,q,2}(x,y) \le 1, \qquad (x,y) \in [0,1]^2.$$

Equality holds exactly at $(0,0)$ *and* $(1,t_0)$. *Consequently every compact subset of* $[0,1]^2$ *disjoint from these two points has a uniform strict gap below one.*

Proof. The estimates used in the proof of Proposition 17.5 give

$$\frac{1}{2} < \Gamma_2(q) \le \frac{3}{2} - \frac{1}{q} < \frac{3}{4},$$

where the first inequality is strict because the interior maximiser beats $t = 0$. Hence $4/3 < p_0 < 2$. By Lemma 5.1, for $s \ge 4/3$,

$$\Gamma_2(s) = \frac{1}{2}\log_2 6 - \frac{1}{s}. \tag{17.17}$$

The unique interior q-edge maximiser strictly beats $t = 1$, so

$$\Gamma_2(q) > \frac{1}{2}\log_2 6 - \frac{1}{q}.$$

Using $1/p_0 = \Gamma_2(q)$ in (17.17) therefore gives $\Gamma_2(p_0) < 1/q$. Thus the edge $x = 1$ has equality only at $y = t_0$, while the edge $y = 1$ is everywhere strict. The axes are strict away from the origin because $p_0, q < 2$ and $c \mapsto c^{-1}\log(1+t^c)$ is strictly decreasing. The boundary reduction Proposition 4.1 gives the inequality on the whole square. Its proof also shows that the logarithm of the reciprocal quotient has no interior local minimum. Hence there can be no additional interior equality point. The compact-gap assertion follows from continuity. □

Theorem 17.8 (Global fixed-q continuation)**.** *Fix $q \in (1, 4/3)$. Let $t_0 = t_q$ be the unique solution of*

$$t_0^{q-1}(2 + t_0) = 1 + 2t_0, \tag{17.18}$$

and put

$$w_0 = -\log t_0, \qquad \alpha_0 = \Gamma_2(q), \qquad p_0 = \frac{1}{\alpha_0}.$$

Define

$$b(q) := \frac{p_0}{4} - \frac{\cosh w_0}{1 + 2\cosh w_0}, \tag{17.19}$$

$$a(q) := \frac{\sinh w_0 \log(2\cosh(w_0/2)) - (w_0/2)\cosh w_0}{2 + 4\cosh w_0}. \tag{17.20}$$

Then $a(q), b(q) > 0$. There are $\varepsilon_q > 0$ and unique real-analytic functions

$$z_{\mathrm{loc}}(r, q), \qquad w_{\mathrm{loc}}(r, q), \qquad \alpha_{\mathrm{loc}}(r, q)$$

for $|r - 2| < \varepsilon_q$, in a neighbourhood of $(0, w_0, \alpha_0)$, such that

$$\Phi = \Phi_z = \Phi_w = 0. \tag{17.21}$$

After possibly decreasing ε_q, for every $r \in [2, 2 + \varepsilon_q)$ one has

$$z_{\mathrm{loc}}(r, q) \geq 0, \qquad w_{\mathrm{loc}}(r, q) > 0, \qquad \alpha_{\mathrm{loc}}(r, q) > 0,$$

and the Hessian in (z, w) is negative definite along the branch. Put

$$p_q(r) := \frac{1}{\alpha_{\mathrm{loc}}(r, q)}.$$

Then

$$\mathcal{R}_{p_q(r),q,r}(x, y) \leq 1, \qquad (x, y) \in [0, 1]^2, \tag{17.22}$$

with equality exactly at

$$(x, y) = (0, 0) \quad \text{and} \quad (x, y) = \left(e^{-z_{\mathrm{loc}}(r,q)}, e^{-w_{\mathrm{loc}}(r,q)}\right). \tag{17.23}$$

Consequently, for arbitrary complex functions supported on Q_d,

$$\|f * g\|_r \leq \|f\|_{p_q(r)} \|g\|_q, \qquad d \geq 1, \tag{17.24}$$

and the sharp constant satisfies

$$\mathsf{C}_d(p_q(r), q; r) = 1.$$

Besides the double point mass, explicit equality cases are the tensor powers of

$$u_r = \left(1, e^{-z_{\mathrm{loc}}(r,q)}\right), \qquad v_r = \left(e^{-w_{\mathrm{loc}}(r,q)}, 1\right). \tag{17.25}$$

No complete classification of equality in dimensions $d > 1$ is asserted.

As $r \to 2$ with q fixed,

$$z_{\mathrm{loc}}(r, q) = \frac{a(q)}{b(q)}(r - 2) + O_q((r - 2)^2), \tag{17.26}$$

$$w_{\mathrm{loc}}(r, q) - w_e(r, q) = O_q((r - 2)^2), \tag{17.27}$$

where $w_e(r, q)$ is the unique global flat-edge maximiser, and

$$\alpha_{\mathrm{loc}}(r, q) = \Gamma_r(q) + \frac{a(q)^2}{2b(q)\log 2}(r - 2)^2 + O_q((r - 2)^3). \tag{17.28}$$

$$p_q(r) = \frac{1}{\Gamma_r(q)} - \frac{a(q)^2}{2b(q)\log 2\, \Gamma_r(q)^2}(r - 2)^2 + O_q((r - 2)^3). \tag{17.29}$$

Every O_q constant and validity radius here may depend on the fixed q. In particular, $z_{\mathrm{loc}}(r,q) > 0$ for $r > 2$ sufficiently close to 2, and the globally sharp biased–biased level requires a strictly larger reciprocal exponent than the edge test.

Proof. At $(z, w, \alpha, r) = (0, w_0, \alpha_0, 2)$, the three equations in (17.21) hold. By Lemma 17.6,

$$\Phi_{zw} = \Phi_{z\alpha} = \Phi_{w\alpha} = 0, \qquad \Phi_\alpha = -\log 2,$$

and

$$\Phi_{zz} = -b(q).$$

To record the other curvature, define

$$K_q(t) := t^{q-1}(2 + t) - (1 + 2t).$$

The one-variable analysis in Lemma 5.1 gives $K_q'(t_0) > 0$, and (17.15) yields

$$c(q) := -\Phi_{ww} = \frac{t_0^2 K_q'(t_0)}{2(1 + t_0 + t_0^2)(1 + t_0^q)} > 0. \tag{17.30}$$

It remains to verify $b(q) > 0$. Return to the balance variables of Section 4 and set

$$\tau_0 := \log\cosh w_0.$$

Since $\log\cosh z = z^2/2 + O(z^4)$ and D is the logarithm of the reciprocal quotient, direct comparison with (17.7) gives

$$b(q) = D_s(0, \tau_0).$$

The quadratic sharp inequality and $D(0, \tau_0) = 0$ imply $b(q) \geq 0$. Suppose that equality held. Edge stationarity would also give $D_\tau(0, \tau_0) = 0$. Put $R = e^{-\tau_0}$ and $h = H''(\tau_0) = R/(2+R)^2$. The two critical equations imply

$$c_{p_0}(0) = c_q(w_0) = \frac{2}{2 + R}, \qquad c_{p_0}(0)(1 - c_{p_0}(0)) = c_q(w_0)(1 - c_q(w_0)) = 2h.$$

Here $p_0, q < 2$. If $\beta = p_0/2$, expansion at the boundary gives

$$A_{p_0}''(0) = \frac{\beta(1 - \beta^2)}{3} < \beta(1 - \beta) = 2h,$$

while (4.11) gives $0 < A_q''(\tau_0) < 2h$. Therefore, if $X = A_{p_0}''(0)/h$ and $Y = A_q''(\tau_0)/h$, then $X, Y \in (0, 2)$ and

$$\frac{\det \nabla^2 D(0, \tau_0)}{h^2} = XY - X - Y = (X - 1)(Y - 1) - 1 < 0.$$

Thus the Hessian has a negative direction with positive s-component. Taylor expansion along a sufficiently short feasible step gives $D < 0$, contradicting the quadratic sharp inequality. Hence $b(q) > 0$.

Thus the Jacobian of (Φ, Φ_z, Φ_w) in (z, w, α) is

$$\begin{pmatrix} 0 & 0 & -\log 2 \\ -b(q) & 0 & 0 \\ 0 & -c(q) & 0 \end{pmatrix}, \tag{17.31}$$

which is invertible. The analytic implicit function theorem proves the branch, and negative definiteness persists from the diagonal Hessian $\operatorname{diag}(-b, -c)$.

Equation (17.16) gives

$$\Phi_{zr}(0, w_0, \alpha_0, q, 2) = a(q).$$

In terms of $t_0 = e^{-w_0}$ its numerator is

$$(1 - t_0^2)\log(1 + t_0) + t_0^2 \log t_0.$$

The strict concavity calculation in the proof of Lemma 17.4 shows that this is positive for $0 < t_0 < 1$, so $a(q) > 0$. Differentiating $\Phi_z = 0$ along the branch and using the vanishing mixed derivatives gives

$$-b(q)\,\partial_r z_{\mathrm{loc}}(2,q) + a(q) = 0,$$

which proves (17.26).

For the second expansion, let $w_e(r,q)$ denote the real-analytic edge maximiser and set $\alpha_e(r,q) = \Gamma_r(q)$. Solve $\Phi_w = 0$ locally for w and substitute it into Φ. With $\epsilon = r-2$ and with the reciprocal exponent fixed at $\alpha_e(r,q)$, the resulting reduced function has the Taylor expansion

$$\widehat{\Phi}(z,\alpha_e,r) = a(q)\epsilon z - \frac{b(q)}{2}z^2 + O_q\big(\epsilon^2|z| + |\epsilon|z^2 + |z|^3\big).$$

Its local maximiser is $z = a(q)\epsilon/b(q) + O_q(\epsilon^2)$, and substitution produces the gain

$$\frac{a(q)^2}{2b(q)}\epsilon^2 + O_q(\epsilon^3).$$

Since $\Phi_\alpha = -\log 2$ at the base point, restoring the level $\Phi = 0$ requires increasing α by this gain divided by $\log 2$. This proves (17.28). Finally, $\Phi_{zw} = \Phi_{w\alpha} = 0$ at $r = 2$, while $z_{\mathrm{loc}} = O_q(\epsilon)$ and $\alpha_{\mathrm{loc}} - \alpha_e = O_q(\epsilon^2)$. The equation $\Phi_w = 0$ and analyticity therefore give $w_{\mathrm{loc}} - w_e = O_q(\epsilon^2)$, so the w adjustment does not alter the displayed coefficient. This is (17.27).

To obtain (17.29), put

$$\delta(r) := \alpha_{\mathrm{loc}}(r,q) - \Gamma_r(q) = \frac{a(q)^2}{2b(q)\log 2}\epsilon^2 + O_q(\epsilon^3).$$

The edge obstruction $\Gamma_r(q)$ is real analytic and stays bounded away from zero. The exact identity

$$\frac{1}{\Gamma+\delta} = \frac{1}{\Gamma} - \frac{\delta}{\Gamma^2} + \frac{\delta^2}{\Gamma^2(\Gamma+\delta)}$$

with $\Gamma = \Gamma_r(q)$ gives the claimed expansion.

It remains to prove the global assertion. For $z, w \ge 0$ define

$$\Psi_r(z,w) := \log \mathcal{R}_{p_q(r),q,r}(e^{-z}, e^{-w}).$$

Multiplying the numerator inside $\mathcal{R}^r$ by $e^{r(z+w)/2}$, and doing the analogous cancellation in the two norm terms, gives the exact identity

$$\Psi_r(z,w) = \Phi_{\alpha_{\mathrm{loc}}(r,q),q,r}(z,w). \tag{17.32}$$

Thus the branch point

$$a_r := \big(z_{\mathrm{loc}}(r,q), w_{\mathrm{loc}}(r,q)\big)$$

has $\Psi_r(a_r) = 0$ and $\nabla\Psi_r(a_r) = 0$. The expansion (17.26) and continuity allow ε_q to be decreased so that $z_{\mathrm{loc}} \ge 0$, $w_{\mathrm{loc}} > 0$, and $\alpha_{\mathrm{loc}} > 0$ on the one-sided interval.

We divide the noncompact quadrant into three regions. First, at $a_0 = (0, w_0)$ the Hessian is $\mathrm{diag}(-b(q), -c(q))$. Hence there are $d_0, \lambda > 0$ such that, after another decrease of ε_q,

$$\nabla^2\Psi_r \le -\lambda I \quad \text{throughout } B(a_0, d_0)$$

for all allowed r, and $a_r \in B(a_0, d_0/2)$. Taylor's formula with integral remainder then gives

$$\Psi_r(u) \le -\frac{\lambda}{2}|u - a_r|^2 \tag{17.33}$$

for $u \in B(a_0, d_0/2) \cap [0,\infty)^2$. Equality in this branch chart occurs only at a_r.

Second, Proposition 17.7 gives $4/3 < p_0 < 2$, and

$$\frac{1}{p_0} + \frac{1}{q} = \Gamma_2(q) + \frac{1}{q} > 1.$$

Choose the final ε_q with its closed one-sided interval inside the analytic domain. By continuity, the triples

$$\big(p_q(r), q, r\big), \qquad 2 \le r \le 2 + \varepsilon_q,$$

form a compact family satisfying

$$1 < p_q(r) < r, \qquad 1 < q < r, \qquad \frac{1}{p_q(r)} + \frac{1}{q} > 1.$$

The tail lemma Lemma 17.2 therefore supplies $\rho_0 > 0$ such that

$$\mathcal{R}_{p_q(r),q,r}(x, y) < 1 \tag{17.34}$$

whenever $(x, y) \neq (0, 0)$ and $\min\{x, y\} < \rho_0$. Decrease ρ_0 so that $\rho_0 < t_0$ and $W_0 := -\log \rho_0 > w_0 + d_0$.

Third, consider the compact logarithmic core

$$C := [0, W_0]^2.$$

At $r = 2$, the equality set in Proposition 17.7 shows that Ψ_2 vanishes in C only at a_0. It is therefore bounded above by some $-\eta < 0$ on the compact set $C \setminus B(a_0, d_0/2)$. Uniform continuity in (r, z, w), after decreasing ε_q once more, gives

$$\Psi_r \le -\frac{\eta}{2} \quad \text{on } C \setminus B(a_0, d_0/2). \tag{17.35}$$

The branch point remains in C. The concave chart (17.33), the compact-middle bound (17.35), and the tail bound (17.34) prove (17.22) and the exact equality set (17.23). Notice in particular that the double point mass $(0, 0)$ persists as an equality case.

We finally pass from the normalized nonnegative ratio to arbitrary complex two-vectors. The coefficientwise triangle inequality gives $|f * g| \le |f| * |g|$. Sort the magnitudes of the two inputs as $A \ge B \ge 0$ and $C \ge D \ge 0$. Reflections do not change any norm, and Lemma 17.4 bounds either relative orientation by

$$U_{\text{opp}} = (AD, AC + BD, BC).$$

For nonzero inputs put $x = B/A$ and $y = D/C$. Homogeneity gives

$$\frac{\|U_{\text{opp}}\|_r}{\|(A, B)\|_{p_q(r)} \|(C, D)\|_q} = \mathcal{R}_{p_q(r),q,r}(x, y) \le 1.$$

This proves the one-dimensional inequality for arbitrary complex inputs. Point masses show that its optimal constant is at least one, hence it is exactly one. Exact tensorisation Proposition 3.1 now proves (17.24) for every d. Finally, convolution and all norms factor for tensor powers; since (u_r, v_r) in (17.25) is the non-point-mass one-dimensional equality pair, its tensor powers give the announced equality examples. $\square$

Corollary 17.9 (Exact constant-one threshold). *In the setting of Theorem 17.8, for every $d \ge 1$, every $r \in [2, 2 + \varepsilon_q)$, and every finite $p \ge 1$,*

$$\mathsf{C}_d(p, q; r) = 1 \quad \Longleftrightarrow \quad 1 \le p \le p_q(r). \tag{17.36}$$

Thus $p_q(r)$ is the exact constant-one threshold in every dimension. If

$$p_e(r, q) := \frac{1}{\Gamma_r(q)}, \qquad \kappa(q) := \frac{a(q)^2}{2b(q) \log 2},$$

then, as $r \to 2$ with q fixed,

$$p_e(r, q) - p_q(r) = \frac{\kappa(q)}{\Gamma_2(q)^2}(r - 2)^2 + O_q((r - 2)^3). \tag{17.37}$$

In particular, $p_q(r) < p_e(r, q)$ for every $r > 2$ sufficiently close to 2.

Proof. If $1 \le p \le p_q(r)$, monotonicity of counting-measure norms gives $\|f\|_p \ge \|f\|_{p_q(r)}$. Hence (17.24) holds with p in place of $p_q(r)$, and point masses show that the optimal constant is one. If $p > p_q(r)$, the first factor u_r in (17.25) has two nonzero coordinates, so strict norm monotonicity gives

$$\|u_r\|_p < \|u_r\|_{p_q(r)}.$$

Since (u_r, v_r) is an equality pair at $p_q(r)$, its quotient at p is strictly larger than one. Tensoring with point masses in the remaining $d-1$ coordinates proves $\mathsf{C}_d(p, q; r) > 1$ for every d.

Finally, (17.29) and analyticity give $\Gamma_r(q) = \Gamma_2(q) + O_q(r-2)$. Subtracting (17.29) from $p_e(r, q) = 1/\Gamma_r(q)$ therefore yields (17.37); its positive leading coefficient gives the strict inequality. □

Corollary 17.10 (Compact-uniform global branch and threshold)**.** *Let $K \Subset (1, 4/3)$ be compact. There are $\varepsilon_K > 0$ and jointly real-analytic functions*

$$z_K(q, r), \quad w_K(q, r), \quad \alpha_K(q, r), \quad p_K(q, r) := \frac{1}{\alpha_K(q, r)}$$

on an open neighbourhood of $K \times [2, 2 + \varepsilon_K]$, agreeing pointwise with the branch in Theorem 17.8, such that, for every $q \in K$, $2 \le r \le 2 + \varepsilon_K$, and $(x, y) \in [0, 1]^2$,

$$\mathcal{R}_{p_K(q,r),q,r}(x, y) \le 1, \tag{17.38}$$

with equality exactly at $(0, 0)$ and

$$(x, y) = \left(e^{-z_K(q,r)}, e^{-w_K(q,r)}\right).$$

For every $d \ge 1$ and every finite $p \ge 1$ one also has, throughout the same parameter rectangle,

$$\mathsf{C}_d(p, q; r) = 1 \quad \Longleftrightarrow \quad 1 \le p \le p_K(q, r). \tag{17.39}$$

Uniformly for $q \in K$ as $r \to 2$,

$$z_K(q, r) = \frac{a(q)}{b(q)}(r-2) + O_K((r-2)^2), \tag{17.40}$$

$$w_K(q, r) - w_e(r, q) = O_K((r-2)^2), \tag{17.41}$$

$$\alpha_K(q, r) = \Gamma_r(q) + \frac{a(q)^2}{2b(q)\log 2}(r-2)^2 + O_K((r-2)^3), \tag{17.42}$$

$$p_K(q, r) = \frac{1}{\Gamma_r(q)} - \frac{a(q)^2}{2b(q)\log 2\,\Gamma_r(q)^2}(r-2)^2 + O_K((r-2)^3). \tag{17.43}$$

The radius and remainder constants may depend on K; no uniformity is asserted as K approaches either endpoint 1 or 4/3.

Proof. Enclose K in a compact interval contained in $(1, 4/3)$. The parameter implicit-function theorem, compactness, and the uniform edge continuation give the jointly analytic branch and the four uniform Taylor remainders. The same three-region decomposition as in Theorem 17.8—now with common concavity, compact-gap, and tail constants—gives (17.38) and its exact equality set with one ε_K. The triangle and orientation inequalities followed by tensorisation give constant one at $p_K(q, r)$ in every dimension. Finally, counting-measure norm monotonicity proves the lower-p half of (17.39), while the positive two-coordinate branch, tensored with point masses, gives a quotient strictly larger than one for $p > p_K(q, r)$. □

Corollary 17.11 (Complete one-dimensional complex equality cases)**.** *In the setting of Theorem 17.8, put*

$$x_q(r) := e^{-z_{\mathrm{loc}}(r,q)}, \qquad y_q(r) := e^{-w_{\mathrm{loc}}(r,q)},$$

let $J(a_0,a_1)=(a_1,a_0)$, *and write* $e_0=(1,0)$, $e_1=(0,1)$. *After decreasing* ε_q,

$$0<y_q(r)<x_q(r)\le 1,$$

with $x_q(r)=1$ *exactly when* $r=2$. *For nonzero complex two-vectors* f,g, *equality in* (17.24) *with* $d=1$ *holds if and only if one of the following occurs.*

(a) *There are* $\lambda,\mu\in\mathbb{C}\setminus\{0\}$ *and* $k,\ell\in\{0,1\}$ *such that*

$$f=\lambda e_k,\qquad g=\mu e_\ell.$$

(b) *There are* $\lambda,\mu\in\mathbb{C}\setminus\{0\}$, $\theta\in\mathbb{R}$, *and* $\sigma\in\{0,1\}$ *such that*

$$f=\lambda J^\sigma\bigl(1,x_q(r)e^{i\theta}\bigr),\qquad g=\mu J^\sigma\bigl(y_q(r)e^{-i\theta},1\bigr). \tag{17.44}$$

Thus all four point-mass support pairs persist, and the two positive-profile copies differ by simultaneous reflection. Swapping the $p_q(r)$*-input with the* q*-input is not an additional equality operation.*

Proof. Sort the coordinate magnitudes of f as $A\ge B\ge 0$ and those of g as $C\ge D\ge 0$, and put $X=B/A$, $Y=D/C$. The proof of Theorem 17.8 gives the chain

$$\|f*g\|_r\le\||f|*|g|\|_r\le\|U_{\mathrm{opp}}\|_r\le\|f\|_{p_q(r)}\|g\|_q.$$

If equality holds, every comparison is equality. The exact normalized equality set (17.23) gives either $(X,Y)=(0,0)$ or $(X,Y)=(x_q(r),y_q(r))$. The first case is precisely (a).

In the second case all four coordinate magnitudes are positive. For $r>2$, the strict part of Lemma 17.4 forces their larger coordinates to be oppositely placed. At $r=2$ the first profile is flat, and the same canonical placement may be chosen. Hence, after a simultaneous reflection, the magnitudes are $A(1,x_q(r))$ and $C(y_q(r),1)$. Equality in the remaining complex triangle inequality is equivalent to f_0g_1 and f_1g_0 having the same argument. Taking $\lambda=f_0$ and $\mu=g_1$ gives a real θ for which this condition is exactly (17.44).

Conversely, point-mass pairs give equality directly. For the family (17.44) with $\sigma=0$, factoring out $\lambda\mu$ leaves convolution

$$\bigl(y_qe^{-i\theta},\,1+x_qy_q,\,x_qe^{i\theta}\bigr),$$

whose quotient is $\mathcal{R}_{p_q(r),q,r}(x_q(r),y_q(r))=1$. Simultaneous reflection reverses the three convolution coordinates, so $\sigma=1$ also gives equality. Finally, $y_q(r)<x_q(r)$ follows at $r=2$ from $t_0<1$ and persists by continuity; $x_q(r)=1$ only at $r=2$ because $z_{\mathrm{loc}}(2,q)=0$ and $z_{\mathrm{loc}}(r,q)>0$ for $r>2$. ∎

Proposition 17.12 (Endpoint degeneration of the fixed-q branch)**.** *Let* $c(q)$ *be defined by* (17.30) *and put*

$$\kappa(q):=\frac{a(q)^2}{2b(q)\log 2}.$$

If $q=1+\varepsilon\downarrow 1$ *and* $\rho_\varepsilon=2^{-1/\varepsilon}$, *then*

$$\begin{aligned}
&t_0=\rho_\varepsilon\Bigl(1+O\Bigl(\frac{\rho_\varepsilon}{\varepsilon}\Bigr)\Bigr), && w_0=\frac{\log 2}{\varepsilon}+O\Bigl(\frac{\rho_\varepsilon}{\varepsilon}\Bigr),\\
&a(q)\sim\frac{t_0}{4}, && b(q)\sim\frac{t_0}{2}, && c(q)\sim\frac{\varepsilon t_0}{2},\\
&\frac{a(q)}{b(q)}\longrightarrow\frac12, && \kappa(q)\sim\frac{t_0}{16\log 2}.
\end{aligned}\tag{17.45}$$

If instead $\delta=4/3-q\downarrow 0$, *define*

$$\alpha_*:=\frac12\log_2 3-\frac14,\qquad p_*:=\frac{1}{\alpha_*},\qquad b_*:=\frac{p_*}{4}-\frac13>0.$$

Then

$$w_0^2 = \frac{81}{2}\delta + O(\delta^2), \qquad a(q) = \frac{\log 2 - 1/2}{6} w_0 + O(w_0^3),$$
$$b(q) = b_* + O(\delta), \qquad c(q) = \frac{\delta}{2} + O(\delta^2). \tag{17.46}$$

In particular, the determinant of the Jacobian in (17.31) is $-(\log 2) b(q) c(q)$ *and tends to zero at both endpoints.*

Proof. The root equation (17.18), with $q = 1 + \varepsilon$, gives

$$\varepsilon \log t_0 = \log(1 + 2t_0) - \log(2 + t_0) = -\log 2 + \frac{3}{2} t_0 + O(t_0^2).$$

It follows first that $t_0/\varepsilon \to 0$ and then, on substituting back, that t_0 and w_0 have the stated asymptotics. The exact identity

$$a(q) = \frac{(1 - t_0^2)\log(1 + t_0) + t_0^2 \log t_0}{4(1 + t_0 + t_0^2)}$$

gives $a(q) \sim t_0/4$. Using $t_0^\varepsilon = (1 + 2t_0)/(2 + t_0)$ in the level formula gives $b(q) \sim t_0/2$, while $K_q'(t_0) \sim \varepsilon/t_0$ in (17.30) gives $c(q) \sim \varepsilon t_0/2$. This proves (17.45).

For the upper endpoint, put $q = 4/3 - \delta$. The logarithmic root equation is

$$-\left(\frac{1}{3} - \delta\right) w_0 = R(w_0), \qquad R(w) := \log(1 + 2e^{-w}) - \log(2 + e^{-w}).$$

Here $R(w) = -w/3 + 2w^3/81 + O(w^5)$, which gives the first formula in (17.46); the formula for $a(q)$ follows by Taylor expansion of (17.20). The level identity gives $p_0 = p_* + O(\delta)$ and hence $b(q) = b_* + O(\delta)$. Finally, apart from an additive constant, the quadratic edge quotient has the expansion

$$F_q(w) = F_q(0) + \frac{\delta}{8} w^2 + \left(-\frac{1}{648} + O(\delta)\right) w^4 + O(w^6).$$

Thus $F_q''(w_0) = -\delta/2 + O(\delta^2)$, and (17.15) gives the claimed asymptotic for $c(q)$. The determinant statement is immediate. ☐

Proposition 17.13 (Exact lower endpoint)**.** *For every* $d \geq 1$ *and every finite* $p, r \geq 1$,

$$\mathsf{C}_d(p, 1; r) = 1 \iff 1 \leq p \leq r, \tag{17.47}$$

and $\mathsf{C}_d(p, 1; r) > 1$ *when* $p > r$*. At the threshold* $p = r$*, every nonzero* f *supported in* Q_d *and every nonzero point mass* g *supported in* Q_d *give equality. Thus the endpoint equality family is non-isolated.*

Proof. Writing convolution as a sum of translates gives

$$f * g = \sum_y g(y) \tau_y f, \qquad \|f * g\|_r \leq \sum_y |g(y)| \, \|\tau_y f\|_r = \|f\|_r \|g\|_1.$$

If $p \leq r$, then $\|f\|_r \leq \|f\|_p$, and point masses show that the resulting constant one is sharp. If $p > r$, take g to be a point mass and f to equal one at two distinct points of Q_d. The quotient is $2^{1/r - 1/p} > 1$. Finally, when $p = r$ and g is any nonzero point mass, convolution is a scalar multiple of a translate of f, so equality holds for every nonzero f. ☐

Remark 17.14. The compact-uniform statement Corollary 17.10 supplies a common r-interval whenever q stays in a fixed compact subset of $(1, 4/3)$. At the upper endpoint, Theorem 17.21 separately closes a fixed neighbourhood of the moving flat transition. At the lower endpoint, Proposition 17.13 gives the exact line $p = r$ at $q = 1$, while Proposition 17.12 shows why the isolated fixed-q branch degenerates as $q \downarrow 1$. No uniform bridge from $q > 1$ to $q = 1$, and no single r-interval valid over the entire q-range, is asserted.

17.3. **The flat–flat transition: local normal form and global closure.** The other end of the quadratic curved boundary has a one-dimensional Hessian degeneracy. Here the fourth-order term decides the local geometry, which will then be globalised in a fixed upper-end parameter neighbourhood.

Proposition 17.15 (Fourth-order expansion at the flat profile)**.** *Put*

$$u = z + w, \qquad v = z - w, \qquad X_r = 2^r, \qquad S_r = 2 + X_r,$$

and

$$A_r = \frac{X_r + 2r}{S_r}, \qquad B_r = \frac{X_r - 2r}{S_r}. \tag{17.48}$$

Then, as $(z, w) \to (0, 0)$*,*

$$\begin{aligned}\Phi_{p,q,r}(z, w) = \Phi_{p,q,r}(0, 0) &+ \frac{(A_r - p)z^2 + 2B_r zw + (A_r - q)w^2}{8} \\ &+ Q_{4,p,q,r}(z, w) + O\big((z^2 + w^2)^3\big),\end{aligned} \tag{17.49}$$

where

$$\begin{aligned}Q_{4,p,q,r}(z, w) := \frac{1}{192}\Bigg\{& p^3 z^4 + q^3 w^4 \\ &+ \frac{r^3(X_r - 4)v^4 + X_r(3r - 2 - X_r)u^4 - 6X_r r^2 u^2 v^2}{S_r^2}\Bigg\}.\end{aligned} \tag{17.50}$$

In particular,

$$4\nabla^2\Phi_{p,q,r}(0, 0) = \begin{pmatrix} A_r - p & B_r \\ B_r & A_r - q \end{pmatrix}. \tag{17.51}$$

Thus a necessary second-order condition for the origin to be a local maximum is

$$p, q \geq A_r, \qquad (p - A_r)(q - A_r) \geq B_r^2. \tag{17.52}$$

If all these inequalities are strict, the origin is a strict local maximum.

Proof. The scalar expansions for cosh and log cosh give

$$N_r(z, w) = S_r + n_2 + n_4 + O\big((z^2 + w^2)^3\big),$$

where

$$n_2 = \frac{r^2}{4}v^2 + \frac{X_r r}{8}u^2, \qquad n_4 = \frac{r^4}{192}v^4 + \frac{X_r r(3r - 2)}{384}u^4.$$

Consequently,

$$\frac{1}{r}\log N_r = \frac{1}{r}\log S_r + \frac{n_2 + n_4}{rS_r} - \frac{n_2^2}{2rS_r^2} + O\big((z^2 + w^2)^3\big).$$

The two norm terms expand as

$$-\frac{1}{p}\log\left(2\cosh\frac{pz}{2}\right) = -\frac{\log 2}{p} - \frac{p}{8}z^2 + \frac{p^3}{192}z^4 + O(z^6),$$

and analogously with q, w. Collecting the quadratic coefficients gives (17.49) through degree two. In degree four, the coefficients of v^4, u^4, u^2v^2 in the numerator logarithm are, respectively,

$$\frac{r^3(X_r - 4)}{192S_r^2}, \qquad \frac{X_r(3r - 2 - X_r)}{192S_r^2}, \qquad -\frac{6X_r r^2}{192S_r^2}.$$

This proves (17.50) and the sixth-order remainder. The Hessian and its definite and semidefinite criteria give the final assertions. ∎

On the full-cube line

$$\frac{1}{p}+\frac{1}{q}=L_r,\qquad L_r:=\frac{1}{r}\log_2(2+2^r),\tag{17.53}$$

the flat profile has level zero. The next theorem treats the simultaneous full-cube and Hessian-degeneracy branch.

Proposition 17.16 (Quadratic upper-end equality and compact gap)**.** *Put*

$$q_*:=\frac{4}{3},\qquad L_2:=\frac{1}{2}\log_2 6,\qquad p_*:=\left(L_2-\frac{3}{4}\right)^{-1}.$$

Then $4/3<p_*<2$ and

$$\mathcal{R}_{p_*,q_*,2}(x,y)\le 1,\qquad (x,y)\in[0,1]^2,\tag{17.54}$$

with equality exactly at $(0,0)$ and $(1,1)$. Consequently, on every compact subset of the square disjoint from those two points there is a uniform strict gap below one.

Proof. The elementary inequalities $6<8$ and $36>32$ give $4/3<p_*<2$. For every $s\ge 4/3$, the proof of (17.17) shows that the quadratic edge objective is strictly increasing on $[0,1]$ and has its unique maximum at $t=1$, with

$$\Gamma_2(s)=L_2-\frac{1}{s}.$$

Thus both finite edges for (p_*,q_*) are sharp only at their common endpoint $(1,1)$, while both axes are strict away from $(0,0)$ because $p_*,q_*<2$. The boundary reduction Proposition 4.1 proves (17.54); its no-interior-minimum curvature argument excludes any additional interior equality point. The compact-gap assertion is then immediate from continuity. ☐

Theorem 17.17 (Analytic flat critical branch)**.** *Let*

$$q_*:=\frac{4}{3},\qquad p_*:=\left(L_2-\frac{3}{4}\right)^{-1}.$$

There are $\varepsilon>0$ and unique real-analytic functions $p_c(r),q_c(r)$, defined for $|r-2|<\varepsilon$ and near (p_*,q_*), such that

$$p_c(2)=p_*,\quad q_c(2)=q_*,\quad \frac{1}{p_c(r)}+\frac{1}{q_c(r)}=L_r,\quad (p_c(r)-A_r)(q_c(r)-A_r)=B_r^2.\tag{17.55}$$

After decreasing ε, for every $r\in[2,2+\varepsilon)$ the origin is a strict local maximum of $\Phi_{p_c(r),q_c(r),r}$ at level zero. Its Hessian is negative semidefinite of rank one. More precisely, if

$$\delta_r:=p_c(r)-A_r,\qquad k_r:=\left(\frac{B_r}{\delta_r},1\right),$$

then $\delta_r>0$, k_r spans the Hessian kernel, and

$$\kappa_r:=Q_{4,p_c(r),q_c(r),r}(k_r)<0.\tag{17.56}$$

At $r=2$,

$$k_2=(0,1),\qquad \kappa_2=Q_{4,p_*,4/3,2}(0,1)=-\frac{1}{648}.$$

Proof. At $r=2$ one has $A_2=4/3$, $B_2=0$, and $L_2=\frac{1}{2}\log_2 6$. Moreover $p_*>4/3$, so $d_*:=p_*-4/3>0$. Define

$$F_1(p,q,r)=\frac{1}{p}+\frac{1}{q}-L_r,\qquad F_2(p,q,r)=(p-A_r)(q-A_r)-B_r^2.$$

At $(p_*,4/3,2)$ their Jacobian in (p,q) is

$$\begin{pmatrix}-p_*^{-2} & -(4/3)^{-2}\\ 0 & d_*\end{pmatrix},$$

whose determinant is nonzero. The analytic implicit function theorem gives the unique branch in (17.55).

Shrink the interval so that $\delta_r > 0$. For $r > 2$ close to 2 one has $B_r > 0$, since $(2^r - 2r)'|_{r=2} = 4\log 2 - 2 > 0$. If $\eta_r = q_c(r) - A_r$, then $\eta_r = B_r^2/\delta_r \geq 0$, and (17.51) becomes

$$\nabla^2\Phi(0,0) = \frac{1}{4}\begin{pmatrix} -\delta_r & B_r \\ B_r & -\eta_r \end{pmatrix}.$$

It has determinant zero, negative trace, and kernel generated by k_r. The first equation in (17.55) gives $\Phi(0,0) = 0$, while central symmetry makes the origin critical.

At the base point, substitution into (17.50) gives

$$Q_{4,p_*,4/3,2}(0,1) = \frac{1}{192}\left[\left(\frac{4}{3}\right)^3 - \frac{8}{3}\right] = -\frac{1}{648}.$$

Continuity gives (17.56) throughout a smaller one-sided interval. To turn this quartic sign into strict local maximality, take $e = (1,0)$ and set

$$G(s,t) := \Phi(sk_r + te).$$

At the origin, $G_{st} = 0$ and $G_{tt} = -\delta_r/4 < 0$. The analytic implicit function theorem solves $G_t = 0$ uniquely as $t = \psi(s)$. Central symmetry and differentiation give $\psi(s) = O(s^3)$. Hence (17.49) gives

$$G(s,\psi(s)) = G(0,0) + \kappa_r s^4 + O(s^6).$$

This is strictly smaller than $G(0,0)$ for small nonzero s. Since $G_{tt} < 0$ in a small rectangle, $t = \psi(s)$ is the unique transverse maximum there. The origin is therefore a strict local maximum. □

Corollary 17.18 (First-order motion of the flat critical branch)**.** *Put $h = \log_2 6$, $\eta = r - 2$, and $d_* = p_* - 4/3$. Then*

$$q_c(r) = \frac{4}{3} + \frac{3 - 2\log 2}{9}\eta + O(\eta^2), \tag{17.57}$$

$$p_c(r) = p_* + p_1\eta + O(\eta^2), \qquad p_1 = -p_*^2\left[\frac{1}{3} - \frac{h}{4} + \frac{3 - 2\log 2}{16}\right], \tag{17.58}$$

$$\frac{B_r}{p_c(r) - A_r} = \frac{2\log 2 - 1}{3d_*}\eta + O(\eta^2). \tag{17.59}$$

In particular, $q_c(r) > 4/3$ for every $r > 2$ sufficiently close to 2.

Proof. Direct differentiation at $r = 2$ gives

$$A_2' = \frac{3 - 2\log 2}{9}, \qquad B_2' = \frac{2\log 2 - 1}{3}, \qquad L_2' = \frac{1}{3} - \frac{h}{4}.$$

Differentiating the determinant identity in (17.55), using $q_c(2) - A_2 = B_2 = 0$ and $p_c(2) - A_2 = d_* > 0$, gives $q_c'(2) = A_2'$. Differentiating the full-cube identity then gives

$$-\frac{p_c'(2)}{p_*^2} - \frac{q_c'(2)}{(4/3)^2} = L_2',$$

which is (17.58). The quotient expansion in (17.59) follows from $B_2 = 0$ and $d_* > 0$. Finally $3 - 2\log 2 > 0$. □

Theorem 17.19 (Supercritical local pitchfork on the full-cube line)**.** *After decreasing the interval in Theorem 17.17, fix $r \geq 2$ in that interval and define, for q near $q_c(r)$,*

$$P_r(q) := \left(L_r - \frac{1}{q}\right)^{-1}, \qquad \mu := q_c(r) - q.$$

There are a neighbourhood U_r of $(0,0)$ and $\nu_r > 0$ such that, whenever $|q - q_c(r)| < \nu_r$ and $p = P_r(q)$, the origin has level zero and is critical, and the following alternatives hold.

(i) *If $q > q_c(r)$, the origin is a nondegenerate strict local maximum and is the only critical point in U_r.*
(ii) *If $q = q_c(r)$, the origin is a degenerate strict local maximum and is the only critical point in U_r.*
(iii) *If $q < q_c(r)$, the origin is a saddle and exactly two other critical points lie in U_r. They are negatives of one another and are nondegenerate strict local maxima with positive Φ value.*

More precisely, put

$$d_r := p_c(r) - A_r, \qquad D(r,q) := (P_r(q) - A_r)(q - A_r) - B_r^2,$$
$$\sigma_r := \frac{\partial_q D(r, q_c(r))}{4d_r}, \qquad \kappa_r := Q_{4,p_c(r),q_c(r),r}(k_r).$$

Then $\sigma_r > 0$, $\kappa_r < 0$, and in coordinates $(z,w) = sk_r + t(1,0)$ the two nonzero points for $\mu > 0$ satisfy

$$s^2 = -\frac{\sigma_r}{4\kappa_r}\mu + O_r(\mu^2), \tag{17.60}$$
$$\Phi = -\frac{\sigma_r^2}{16\kappa_r}\mu^2 + O_r(\mu^3). \tag{17.61}$$

At $r = 2$, $\sigma_2 = 1/4$ and $\kappa_2 = -1/648$, so

$$s^2 = \frac{81}{2}\left(\frac{4}{3} - q\right) + O\left(\left(\frac{4}{3} - q\right)^2\right), \qquad \Phi = \frac{81}{32}\left(\frac{4}{3} - q\right)^2 + O\left(\left(\frac{4}{3} - q\right)^3\right).$$

Proof. The definition of P_r gives full-cube equality, hence $\Phi(0,0) = 0$; central symmetry makes the origin critical. Write

$$d(r,q) = P_r(q) - A_r, \qquad e(r,q) = q - A_r.$$

Then (17.51) gives

$$H(r,q) = \frac{1}{4}\begin{pmatrix} -d(r,q) & B_r \\ B_r & -e(r,q) \end{pmatrix}, \qquad \det(4H) = D(r,q).$$

Since $P_r'(q) = -P_r(q)^2/q^2$,

$$\partial_q D = P_r'(q)e + d.$$

At $(r,q) = (2, 4/3)$ this equals $p_* - 4/3 > 0$, and it remains positive along a smaller transition interval. Hence $\sigma_r > 0$.

Set $e_1 = (1,0)$ and

$$G_r(s,t,\mu) := \Phi_{P_r(q_c(r)-\mu),\, q_c(r)-\mu,\, r}(sk_r + te_1).$$

At the transition, $\partial_{tt}G_r = -d_r/4 < 0$, so the analytic implicit function theorem solves $\partial_t G_r = 0$ uniquely as $t = \psi_r(s,\mu)$. Symmetry makes ψ_r odd in s and the reduced function

$$g_r(s,\mu) := G_r(s, \psi_r(s,\mu), \mu)$$

even in s. At $\mu = 0$, the proof of Theorem 17.17 gives

$$g_r(s,0) = \kappa_r s^4 + O_r(s^6).$$

The reduced second derivative is the Schur complement of the transverse entry. Since the coordinate matrix with columns k_r, e_1 has determinant -1,

$$\partial_{ss} g_r(0,\mu) = -\frac{D(r, q_c(r) - \mu)}{4d(r, q_c(r) - \mu)} = \sigma_r \mu + O_r(\mu^2).$$

Analyticity and evenness therefore give

$$g_r(s,\mu) = \frac{\sigma_r}{2}\mu s^2 + \kappa_r s^4 + O_r\big(\mu^2 s^2 + |\mu| s^4 + s^6\big). \tag{17.62}$$

Factoring $\partial_s g_r = sR_r(s^2,\mu)$ and applying the implicit function theorem in $u = s^2$ gives the unique nonzero squared amplitude

$$u_r(\mu) = -\frac{\sigma_r}{4\kappa_r}\mu + O_r(\mu^2).$$

It is positive exactly when $\mu > 0$. At the two square roots the reduced second derivative is negative, and the transverse second derivative remains negative, so their full Hessians are negative definite. Substitution into (17.62) proves (17.61). At the origin, the determinant changes sign with $q - q_c(r)$, giving the stated maximum and saddle alternatives; the transition case is Theorem 17.17. The implicit-function uniqueness statements give the exact local critical-point counts after shrinking U_r, ν_r. Finally, substituting $\sigma_2 = 1/4$ and $\kappa_2 = -1/648$ gives the displayed quadratic endpoint constants. □

Corollary 17.20 (Uniform local zero-level surface near the flat transition)**.** *There are $\varepsilon_0, \delta_0 > 0$ and jointly real-analytic functions $u_*(r,\mu), \lambda_*(r,\mu)$ for $|r-2| < \varepsilon_0$, $|\mu| < \delta_0$, with $u_*(r,0) = \lambda_*(r,0) = 0$, as follows. Put*

$$q = q_c(r) - \mu, \qquad \alpha_{\text{flat}}(r,q) := L_r - \frac{1}{q},$$

and use σ_r, κ_r from Theorem 17.19. Uniformly for r near 2,

$$u_*(r,\mu) = -\frac{\sigma_r}{4\kappa_r}\mu + O(\mu^2), \tag{17.63}$$

$$\lambda_*(r,\mu) = -\frac{\sigma_r^2}{16\kappa_r \log 2}\mu^2 + O(\mu^3). \tag{17.64}$$

Both are positive for small $\mu > 0$. At $r = 2$ their leading coefficients are $81/2$ and $81/(32\log 2)$, respectively.

Moreover, there is one fixed neighbourhood U of the flat profile such that, after decreasing ε_0, δ_0, the following alternatives hold for $2 \le r < 2 + \varepsilon_0$.

(i) *If $-\delta_0 < \mu \le 0$ and $\alpha = \alpha_{\text{flat}}(r,q)$, then $\Phi_{\alpha,q,r} \le 0$ on U, with equality only at $(0,0)$.*

(ii) *If $0 < \mu < \delta_0$ and $\alpha = \alpha_{\text{flat}}(r,q) + \lambda_*(r,\mu)$, then $\Phi_{\alpha,q,r} \le 0$ on U, with equality exactly at two nonzero points. They are negatives of one another, have squared kernel amplitude $u_*(r,\mu)$, and have negative-definite (z,w) Hessian.*

Proof. Let $e_1 = (1,0)$ and set

$$G(s,t,r,\mu,\lambda) := \Phi_{\alpha_{\text{flat}}(r,q)+\lambda,q,r}(sk_r + te_1), \qquad q = q_c(r) - \mu.$$

The transverse second derivative is strictly negative, uniformly for r near 2. The parameter implicit function theorem therefore solves $G_t = 0$ by a jointly analytic graph $t = \psi(s,r,\mu,\lambda)$, odd in s. The reduced function is even and has the form

$$G(s,\psi(s,r,\mu,\lambda),r,\mu,\lambda) = F(u,r,\mu,\lambda), \qquad u = s^2,$$

where

$$F(0,r,\mu,\lambda) = -\lambda\log 2$$

and, at $\lambda = 0$,

$$F(u,r,\mu,0) = \frac{\sigma_r}{2}\mu u + \kappa_r u^2 + O(\mu^2 u + |\mu|u^2 + u^3),$$

uniformly for r near 2. The Jacobian of $F_u = F = 0$ in (u,λ) at $(u,\mu,\lambda) = (0,0,0)$ is

$$\begin{pmatrix} 2\kappa_r & F_{u\lambda}(0,r,0,0) \\ 0 & -\log 2 \end{pmatrix},$$

which is uniformly invertible. The joint analytic implicit function theorem gives u_*, λ_*. Differentiation and substitution give (17.63)–(17.64); their signs and the endpoint coefficients follow from $\sigma_r > 0 > \kappa_r$, $\sigma_2 = 1/4$, and $\kappa_2 = -1/648$.

Shrink a common coordinate box so that $G_{tt} < 0$ and $F_{uu} < 0$ throughout. For $\mu \le 0$ and $\lambda = 0$, the Schur-complement identity from Theorem 17.19 gives

$$2F_u(0, r, \mu, 0) = -\frac{D(r, q_c(r) - \mu)}{4\big(P_r(q_c(r) - \mu) - A_r\big)} \le 0.$$

Strict concavity in $u \ge 0$ and $F(0, r, \mu, 0) = 0$ give alternative (i). For $\mu > 0$, the defining equations give $F(u_*, r, \mu, \lambda_*) = F_u(u_*, r, \mu, \lambda_*) = 0$; strict concavity makes u_* the unique reduced maximiser. The two square roots of u_* and strict transverse concavity give exactly the two equality points and their negative-definite Hessians, proving (ii). $\square$

Theorem 17.21 (Global upper-end flat-to-biased transition)**.** *There are $\varepsilon, \delta > 0$ such that the following holds whenever*

$$2 \le r < 2 + \varepsilon, \qquad |\mu| < \delta, \qquad q = q_c(r) - \mu.$$

Put

$$\alpha_{\text{flat}}(r, q) := L_r - \frac{1}{q},$$

and define the sharp reciprocal first exponent by

$$\alpha_{\text{sh}}(r, \mu) := \begin{cases} \alpha_{\text{flat}}(r, q), & \mu \le 0, \\ \alpha_{\text{flat}}(r, q) + \lambda_*(r, \mu), & \mu > 0, \end{cases} \qquad p_{\text{sh}}(r, \mu) := \frac{1}{\alpha_{\text{sh}}(r, \mu)}. \tag{17.65}$$

The function α_{sh} is continuous and is real analytic on each side of $\mu = 0$. For every $(x, y) \in [0, 1]^2$,

$$\mathcal{R}_{p_{\text{sh}}(r,\mu),q,r}(x, y) \le 1. \tag{17.66}$$

The exact normalized equality set is as follows.

(i) *If $\mu \le 0$, equality holds exactly at $(0, 0)$ and $(1, 1)$. In particular, the transition value $\mu = 0$ belongs to the flat phase.*

(ii) *If $\mu > 0$, equality holds exactly at $(0, 0)$ and one finite point*

$$(x, y) = \left(e^{-z_+(r,\mu)}, e^{-w_+(r,\mu)}\right), \qquad w_+(r, \mu) := \sqrt{u_*(r, \mu)} > 0. \tag{17.67}$$

Here $z_+(r, \mu) \ge 0$, $z_+(2, \mu) = 0$, and $z_+(r, \mu) > 0$ when $r > 2$.

Thus the finite equality profile is flat–flat for $\mu \le 0$, flat–biased when $r = 2$ and $\mu > 0$, and biased–biased when $r > 2$ and $\mu > 0$.

Proof. The identity (17.32), with the parameters changed, gives

$$\log \mathcal{R}_{1/\alpha,q,r}(e^{-z}, e^{-w}) = \Phi_{\alpha,q,r}(z, w). \tag{17.68}$$

Since $\lambda_*(r, 0) = 0$, the piecewise function (17.65) is continuous. At $(r, \mu) = (2, 0)$ it gives $(p, q) = (p_*, 4/3)$ from Proposition 17.16. After shrinking the parameter rectangle, all triples $(p_{\text{sh}}(r, \mu), q, r)$ lie in a compact set satisfying

$$1 < p_{\text{sh}} < r, \qquad 1 < q < r, \qquad \frac{1}{p_{\text{sh}}} + \frac{1}{q} > 1.$$

The tail lemma Lemma 17.2 therefore gives $\rho > 0$ such that the ratio is strictly below one at every nonorigin point with $\min\{x, y\} < \rho$; the corner $(0, 0)$ itself always has equality.

It remains to control the compact logarithmic core $C = [0, -\log \rho]^2$. At the base parameter, Proposition 17.16 says that the only equality point in C is $(z, w) = (0, 0)$. Let U be the fixed local neighbourhood in Corollary 17.20, and choose a smaller symmetric neighbourhood V with $\overline{V} \subset U$. The base logarithmic quotient has a strict negative maximum on $C \setminus V$. Uniform continuity in (r, μ, z, w), using continuity of α_{sh}, preserves half of this gap after another shrink. Inside V, the two alternatives in Corollary 17.20 give the required inequality and its local equality set. Together with the tail estimate, these bounds prove (17.66) globally.

For $\mu \le 0$, the sole finite logarithmic equality is the origin, hence the finite normalized equality is $(1,1)$. Suppose $\mu > 0$. In the coordinates $(z,w) = sk_r + t(1,0)$ used above, the positive soft root $s = \sqrt{u_*(r,\mu)}$ has $w = s > 0$; denote its first coordinate by z_+. If $r = 2$, then $k_2 = (0,1)$ and Lemma 17.6 gives $\Phi_z(0,w) = 0$ for every w. Uniqueness of the transverse critical graph forces $z_+(2,\mu) = 0$. If $r > 2$, then Proposition 17.5 excludes $z_+ = 0$. Nor can $z_+ < 0$: the strict orientation inequality Lemma 17.4 would make $\Phi(-z_+, w_+) > \Phi(z_+, w_+) = 0$, contradicting the local bound in the symmetric neighbourhood U. Hence $z_+ > 0$. The other local equality point is $(-z_+, -w_+)$ and lies outside the quadrant. This proves the exact equality classification. $\square$

Corollary 17.22 (All-dimensional upper-transition threshold)**.** *In the parameter neighbourhood of Theorem 17.21, for every $d \ge 1$ and every finite $p \ge 1$,*

$$\mathsf{C}_d(p,q;r) = 1 \iff \frac{1}{p} \ge \alpha_{\mathrm{sh}}(r,\mu) \iff p \le p_{\mathrm{sh}}(r,\mu), \tag{17.69}$$

and the constant is strictly larger than one when $1/p < \alpha_{\mathrm{sh}}(r,\mu)$. At the threshold, tensor powers of

$$u_{\mathrm{fl}} = v_{\mathrm{fl}} = (1,1)$$

are equality cases when $\mu \le 0$. When $\mu > 0$, tensor powers of

$$u_+ = \left(1, e^{-z_+(r,\mu)}\right), \qquad v_+ = \left(e^{-w_+(r,\mu)}, 1\right) \tag{17.70}$$

are equality cases. No complete classification of equality in dimensions $d > 1$ is asserted.

Proof. At $p = p_{\mathrm{sh}}$, the triangle inequality and Lemma 17.4 and Theorem 17.21 give the inequality for arbitrary complex two-vectors. Point masses give sharpness, and exact tensorisation Proposition 3.1 gives constant one in every dimension. For $p < p_{\mathrm{sh}}$, counting-measure norm monotonicity preserves the inequality. For $p > p_{\mathrm{sh}}$, the relevant finite threshold equality factor—u_{fl} or u_+—has two nonzero coordinates, so strict norm monotonicity makes its quotient strictly larger than one. Tensoring with point masses proves the same in every dimension. Finally, convolution and all norms factor on tensor powers, giving the displayed equality examples. $\square$

Remark 17.23. The words *local* and *in U_r* in Theorems 17.17 and 17.19 are essential. These results and Corollary 17.20 determine the local normal form; Theorem 17.21 and Corollary 17.22 combine it with compact-gap and tail arguments to close the upper-end neighbourhood globally. This does not produce one r-radius uniform over the entire q-range. The endpoint $q = 1$ is exact by Proposition 17.13, but a uniform bridge from $q > 1$ down to that endpoint remains open.

18. Concluding perspective

The three alphabet regimes display a common hierarchy. In the binary model, product structure and the Hilbertian output reduce the sharp problem to boundary geometry. In the ternary model, an interior but still one-parameter symmetric branch becomes extremal, and the binary estimate survives as one half of a two-envelope argument. For larger alphabets, no finite list of balance parameters is expected to control the global problem. The exact replacement is multiscale: local parity gains, nonnegative defects, and pathwise products.

Two principal problems remain. First, the refinement-rigidity conjecture must be proved in order to derive the sharp Gaussian first-order asymptotic for growing alphabets. Second, away from $r = 2$, the complete off-diagonal binary phase diagram remains open beyond the local regimes closed in the transverse epilogue. The exact small models suggest that both questions should be approached through the geometry of equality and near equality rather than through a direct global optimization in all coordinates.

Appendix A. Certified numerical enclosures

Only a finite list of numerical comparisons is needed in the exact ternary arguments. The decimal endpoints below are terminating decimals and may be verified by rational interval arithmetic using the standard expansion

$$\log y = 2\sum_{j=0}^{M-1} \frac{z^{2j+1}}{2j+1} + R_M, \qquad z = \frac{y-1}{y+1},$$

with

$$0 < R_M < \frac{2z^{2M+1}}{(2M+1)(1-z^2)}$$

after reduction of the logarithm argument by a power of two.

For the critical root function,

$$\begin{aligned} \Psi(3) &\in [0.00213685582205790, 0.00213685582205791], \\ \Psi(3.1) &\in [-0.00099534379472778, -0.00099534379472776], \\ \Psi'([3, 3.1]) &\subset [-0.050, -0.012]. \end{aligned}$$

Consequently,

$$\begin{aligned} x_* &\in [3.06689294695624189, 3.06689294695624190], \\ p_* &\in [1.470203929717889497, 1.470203929717889506], \\ t_* &\in [2.720710997397171373, 2.720710997397171389]. \end{aligned}$$

The overlap comparisons use

$$\begin{aligned} \alpha_{1.477}(1/3) &\in [4.52720013215865, 4.52720013215867], \\ 2^{4/1.477} - 2 &\in [4.53502285532090, 4.53502285532092], \\ \alpha_{1.467}(1) + \beta_{1.467}(1) - 1 &\in [4.57067, 4.57069]. \end{aligned}$$

The final interval in the original two-envelope proof is

$$\frac{59}{243}\left(\frac{2}{1+3^{-1.467}}\right)^{4/1.467} \in [0.97856411604898, 0.97856411604899].$$

These intervals have substantial margins relative to all strict comparisons used above.

Statement on the use of AI

AI systems were used during exploratory work and preparation, including parametrization, symbolic and numerical checks, literature and file searches, proof stress testing, exposition, reorganization of the component manuscripts, and LaTeX preparation. The author retains responsibility for the final mathematical and expository decisions and for verifying the manuscript before submission.

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